# Politicians from 12 countries rarely engage with researchers on social media, but this can change when expertise gains salience

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**Abstract.** Interactions between the policy and academic communities can play an important role in political decisionmaking. Still, the fact that much of the policymaking process happens behind closed doors obscures our understanding of the relationships between political decision-makers with academic researchers. To address this challenge, this paper analyzes online behavioral data to examine otherwise hidden interactions between 3,670 lawmakers in 12 countries with 410K academic researchers on Twitter. The findings suggest that lawmakers do follow, yet rarely visibly engage with researchers online. Lawmakers from conservative and radical right parties, whose electoral coalitions tend to be more skeptical of expert authority, follow and engage less with researchers online than their colleagues from other parties. While the base engagement is relatively low across legislatures, it can increase when expertise gains salience, suggesting that the constraints on publicly visible engagement with researchers are context-dependent rather than fixed. During the early stages of the COVID-19 pandemic, marked by policy uncertainty involving a novel and technically complex policy issue, lawmakers' overall inclination to follow and engage with scholars increased, most prominently targeting researchers from the medical sciences. These findings offer new insights into when and how lawmakers publicly attend to academic expertise, contributing to a broader understanding of political elites' symbolic and informational engagement with science.

**Keywords.** legislators | politics of expert knowledge | elite digital traces | social media

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Global challenges such as climate change and the COVID-19 pandemic underscore the potential importance of academic researchers, scientific research and insights on policy formation (Bavel et al., 2020; Berger et al., 2021). Arguably, research evidence can be valuable for policy decisions by presenting information to create and execute policies to address societal problems. At the same time, we live in a period with an unprecedented expansion of the scientific workforce, accompanied by a surge in research output (Heuer, Einaudi, and Kang, 2023; Fortunato et al., 2018; Bornmann, Haunschild, and Mutz, 2021). The increasingly growing pool of academic experts and knowledge converges with the reality that policymakers receive a steady flow of information on societal issues from diverse sources (Jones and Baumgartner, 2005; Senninger and Seeberg, 2024). Consequently, the decisions policymakers make about engaging with researchers and evidence—and ultimately whom they pay attention to—can have significant implications.

Yet engaging with academic research, and researchers, is not a costless act, even more for elected officials. Legislators operate in electoral environments in which their public behavior sends signals to constituents, activists, party elites, and media audiences. Visible engagement with academic researchers—discussing their work, responding to their claims, or publicly amplifying their contributions on social media—can signal an epistemic orientation that values specialized knowledge and scientific authority, as well as an association with academic and educational institutions. For legislators whose electoral coalitions trust such institutions, these signals may be benign or even beneficial. For those representing constituencies skeptical intellectual elites, visible engagement with academics may entail political risks. This suggests that legislators' relationships with researchers could be shaped not only by their informational needs but also by the audience costs associated with being seen to seek or endorse academic expertise.

There is a long-standing research tradition in the social sciences looking at the interactions between expert academic communities and public decision-makers (Weiss, 1979; Caplan, 1979; Huberman, 1994; Haas, 1992). However, there are gaps in our understanding about whether actors in the policy sphere engage with researchers and are exposed to and consume evidence (National Research Council, 2012). One of the main obstacles to studying this is the confidentiality and unobservability of elite behaviors. Much of the previous work in this domain relies on participant observation and semi-structured interviews (Geddes, 2021; Bogenschneider, Day, and Bogenschneider, 2021), self-reports (Caplan et al., 1975; Weiss and Bucuvalas, 1980), and information provision experiments (Vivalt and Coville, 2023; Lee, 2022; Baekgaard et al., 2019). While these approaches have deepened our understanding of how policymakers encounter and interpret expertise, they provide limited leverage on the public-facing dimension of engagement, precisely the arena in which signaling and audience costs should be most consequential.

In this paper, I propose a complementary lens that uses digital behavioral data from social media accounts to observe publicly visible engagement with academic researchers. Social media platforms allow legislators to follow, reference, and interact with experts in ways that are observable to broad audiences. These behaviors are therefore not merely informational acts, but also public signals that may reflect legislators' calculations about expertise, legitimacy, and representation. I investigate the legislator relationships with academic researchers on Twitter (now X) and explore potential variation across different political contexts, jurisdictions, and individuals. I suggest that online engagement captures a visible margin of publicly observable contact with academic researchers that is shaped by both legislators' underlying demand for information and the political costs of publicly signaling engagement with academic knowledge. As a result, engagement should vary systematically with legislators' ideological positioning and with contextual shifts in the salience of expertise.

Equipped with granular social media data from legislators across 12 democracies, I tackle three overarching sequential questions: 1. Do we observe lawmakers following and engaging with researchers 'in the wild'? 2. Are there legislator and legislature-level factors that help predict online engagement with researchers? And, 3. do legislators adapt their public engagement behaviors when demand for academic expertise increases?

The analyses show that (i) academic researchers are present in legislators' social media networks, but that visible encounters between legislators and researchers are rare; (ii) engagement varies systematically across legislators, with political affiliation and educational background playing an important role: conservative and radical-right legislators are significantly less likely to follow or interact publicly with academic researchers; and (iii) engagement is not fixed. During periods of heightened demand for expert knowledge, such as the COVID-19 pandemic, legislators across contexts become more likely to follow and engage with domain-relevant researchers. These patterns are consistent with a signaling account in which baseline engagement is constrained by audience costs that vary across ideological coalitions, but in which those constraints can ease when the salience and electoral value of expertise increase.

### Studying political and scientific elites on Twitter

Online social media platforms are increasingly important around the globe. These media have transformed market practices, information diffusion and consumption, and political communication (Lamberton and Stephen, 2016; Westerman, Spence, and Van Der Heide, 2014; Jungherr, 2016). These platforms also hold particular relevance for both political and scientific communication, offering actors in both spheres new ways to connect with broader publics and each other (Brainard, 2022).

A recent global study across 68 countries finds that social media are now the most significant source of science-related information in most regions (Mede, Cologna, et al., 2024). If this is where much of the public encounters scientific knowledge, it becomes increasingly timely to ask whether lawmakers also encounter, and publicly engage with, researchers in these spaces. Unlike traditional elite communications, which are often private, curated, or constrained, social media offer a comparatively open and observable venue where selective forms of engagement can be systematically studied.

For lawmakers, social media represent a low-cost channel to convey information to the public and appeal to constituents. These media provide flexibility to communicate, receive feedback, and gather information without the constraints of other political arenas, such as regulated debates, prepared statements, and news coverage (Castanho Silva and Proksch, 2022). For academics, social media represent a tool to advance their careers, expand knowledge networks, and scale research dissemination to reach a wider audience (Garg and Fetzer, 2025; Klar et al., 2020).

Crucially, interactions on social media produce traces that enable the large-scale analysis of public-facing behaviors. Prior research has shown that online networks often reflect offline social structures, and that digital behaviors can meaningfully reveal underlying 'analogue' dispositions (Dunbar et al., 2015; Barberá, 2015; He and Tsvetkova, 2023). In the context of political elites, such data have been used to study how legislators interact with constituents (Spierings, Jacobs, and Linders, 2019) and interest groups (Bunea, Ibenskas, and Weiler, 2025). The underlying logic in this work is that decisions to follow, reply to, or mention others on platforms like Twitter signal attention, interest, or perceived informational relevance.

Building on this logic, I argue that legislators' engagement with researchers online can be interpreted as public signals of attention to academic actors. These behaviors do not constitute direct evidence of evidence use or policy learning, but they offer observable indicators of legislators' willingness to acknowledge, amplify, or associate with scientific expertise in the public sphere. Importantly, the public nature of this engagement is not incidental—it is analytically consequential. Unlike private consultations or committee briefings, social media interactions are visible to constituents, party elites, journalists, and broader audiences. A legislator who follows, retweets, or mentions a researcher is not merely consuming information; they are making a visible choice that can be read as an endorsement of a particular epistemic authority.

This visibility implies that engagement with researchers can carry audience costs. For legislators whose electoral coalitions are broadly supportive of scientific institutions and expert authority, publicly engaging with researchers may be expected and even electorally beneficial. For others, particularly those representing constituencies skeptical of expert elites or aligned with political movements that mobilize anti-expert sentiment, publicly engaging with researchers may risk alienating important audiences. This logic would predict systematic variation in engagement along ideological lines: for instance, legislators from conservative and radical right parties, whose electoral coalitions are more likely to harbor skepticism toward academic expertise (Mede and Schäfer, 2020), should exhibit lower baseline rates of publicly visible engagement with researchers. By contrast, legislators from green parties, whose coalitions tend to value scientific knowledge and environmental expertise, should engage more. This reasoning does not require that legislators consciously weigh each interaction as a strategic decision; it simply suggests that the broader political context can shape the incentive structure within which engagement occurs.

Audience costs, however, are not fixed. They can be modulated by changes in the political environment, in particular, by shifts in the salience of expertise. When a policy challenge is widely perceived as technically complex, urgent, and consequential, expert communities with specialized knowledge gain salience (Haas, 1992; Dunlop, 2017). This can mean that the electoral value of being seen to consult and amplify expert knowledge may increase, even for legislators who would otherwise avoid such visible associations. Under conditions of high epistemic demand, engaging with researchers might become less politically costly and potentially more rewarding as a signal of competence and seriousness. This suggests that the contextual salience of expertise can be an important factor in engagement: baseline constraints can ease when the demand for expert knowledge becomes politically legible. Crises such as a global pandemic constitute a particularly clear instance of this dynamic. The COVID-19 pandemic generated acute uncertainty about novel and technically complex policy problems, and created conditions under which being seen to engage with relevant scientific expertise became broadly valued across political contexts.

Under the signaling account, we could expect an increase in legislators' visible engagement with researchers at the onset of the pandemic—concentrated on researchers with expertise directly relevant to the crisis.

This online setting also offers a unique opportunity to compare patterns of elite–expert engagement across contexts. Legislator accounts operate within a unique framework of platform-imposed behaviors and embed all users within a shared public network, making the behavioral signals comparable. Moreover, different types of interactions on platforms like Twitter carry distinct social meanings—varying in terms of visibility, cognitive cost, and symbolic weight (Metaxas et al., 2015; Wojcieszak et al., 2022). Examining these behaviors can shed light on the strategic, partisan, and situational factors that shape when and how political elites publicly signal engagement with scientific expertise.

Furthermore, Twitter represents a particularly valuable setting for this line of inquiry. Unlike other platforms, it attracted disproportionately high concentrations of both political elites and academic researchers, making it one of the few digital spaces where these two communities genuinely cohabited. The fact that both groups operated within a shared, observable public network is precisely what makes the signaling logic legible: legislators and researchers were not communicating in separate spheres but in a common arena where associations, endorsements, and silences were mutually visible. The pre-2022 period represents a bounded and well-documented window of elite digital public life that is unlikely to be replicated on successor platforms.

## Data, methods, and empirical setting

The data for this study were collected between November and December 2022 and center on the Twitter activity of elected legislators in office during that year. The dataset spans 12 democracies with widespread Twitter adoption, including the three largest in both South and North America (Argentina, Brazil, Colombia, Mexico, Canada, and the United States), as well as six in Europe (France, Germany, Ireland, Italy, Spain, and the United Kingdom). This cross-national scope provides a broad comparative foundation for analyzing patterns of interaction between legislators and academic researchers on social media.

To examine the relationship of legislators with academic researchers on social media and answer the three overarching questions, I match an original dataset containing the historical Twitter behavior of 3,670 legislators from 12 different countries with a newly compiled database of academic researchers on Twitter.

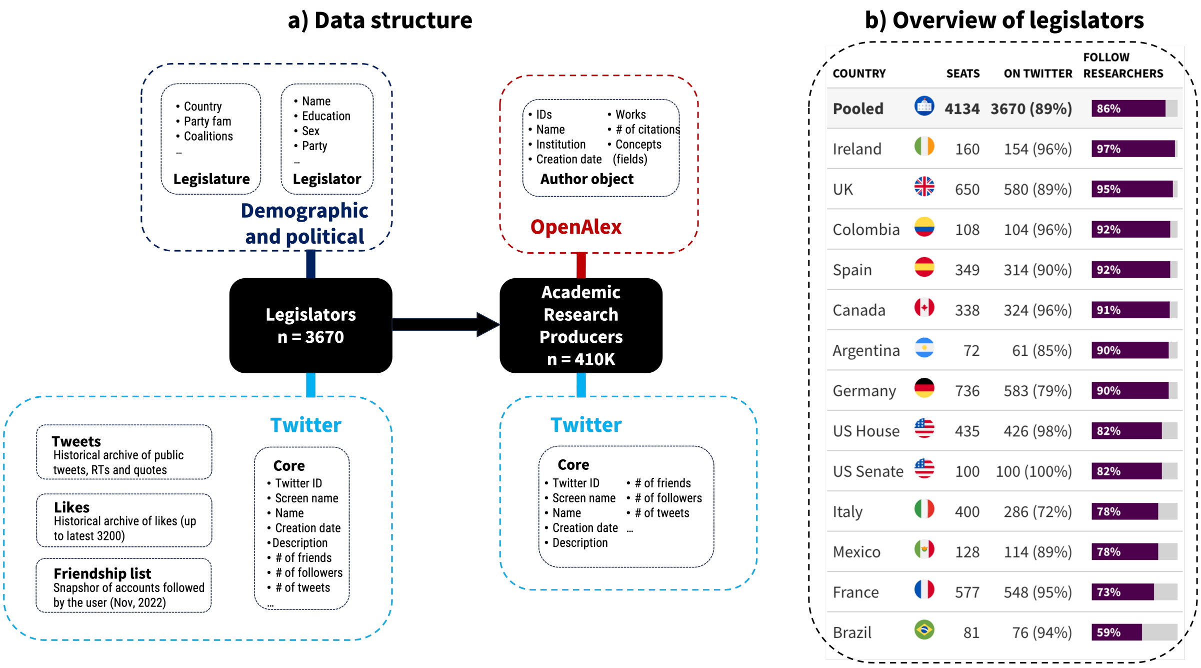


**Figure 1: Study population and data structure.** Panel a presents an overview of the data structure and connections between the legislator and academic researcher data sources. Panel b provides a summary of the legislators included in the study.

On the legislator-end, I collect all public posts (≈20M), a list of their followed accounts (≈2.6M), and the historical archive of liked posts (≈6.5M) by lawmakers from 12 democracies in Western Europe, North and South America actively in office during 2022. Additionally, I supplement the Twitter data with legislators' demographic and political information.

On the academic researcher-end, the starting point is a public database of 410K researcher Twitter and OpenAlex IDs (Mongeon, Bowman, and Costas, 2023). These accounts were identified by Mongeon et al. using a high-precision matching approach that connects Twitter profiles to researchers who shared links to scholarly work on Twitter, based on data from the Crossref Event Data release (January 2022). I collect information from their Twitter profiles, linking them to their respective researcher entries on the open index of scholarly work, OpenAlex (Priem, Piwowar, and Orr, 2022). I extract background information about the academics, such as their scientific field.

Figure 1 provides an overview of the study population and data structure. All-in-all, roughly 90% of the legislators in these democracies were on Twitter. For further information and methodological detail about the data collection, see *SI Appendix A*.

I employ the list of researcher accounts to explore the instances where legislators engage with researchers in the platform. I contribute to past work on the relationship between academic researchers, evidence, and legislators in three ways.

First, I employ a set of behavioral measures to gauge legislators' engagement with academic researchers in the digital realm. Traditionally, studies in this area have relied on self-reported measures (Avey and Desch, 2014; Riphahn and Schnitzer, 2022; Seidel et al., 2021). While asking MPs how much they engage with academic researchers in their parliamentarian activities is valuable (Dodson, Geary, and Brownson, 2015; Purtle et al., 2018), these measures can be costly and difficult to scale beyond one context.

I study six specific behaviors, each giving different affordances to users and shedding light on diverse facets of the Twitter interaction landscape:

- **Following:** This behavior involves users subscribing to the updates of other accounts on the platform, reflecting their interest in specific users.
- **Mentioning:** This is the behavior by which users reference an account in their original posts. It entails direct engagement and communication with that user.
- **Retweeting:** This action entails users forwarding posts made by other accounts to their own followers, potentially amplifying the reach of specific content.
- **Quote Tweeting:** This function allows users to share another account's post while adding their commentary, potentially contributing their perspective to the content.
- **Replying:** When a user responds to posts made by other users. It can reflect direct engagement in conversations and discussions taking place on the platform.
- **Liking:** By using the 'like' function, users generally express positive sentiments or approval for particular tweets, potentially contributing to the overall sentiment and visibility of content.

These behaviors provide indicators that can help understand the interactions between legislators and researchers on Twitter.

Second, leveraging the scalability of these measures, I extend the analyses to lawmakers from 12 countries. The majority of previous studies exploring research evidence in legislatures present insights from single-country studies, with a lopsided focus on the U.S. context (Ouimet et al., 2023). This international perspective enables us to better understand the connections between legislators and academic research providers across diverse political, institutional, and cultural contexts. I include legislators from various countries to examine correlates at the individual and legislature levels, such as legislators' professional backgrounds, political party affiliations, and political system characteristics. As such, I assess patterns and factors related to engagement with researchers across country contexts.

Third, I examine the responsiveness of legislators' engagement to changes in the salience of expertise. The COVID-19 pandemic serves as an intriguing case study. The early stages of the pandemic generated acute policy uncertainty around a novel and technically complex issue, creating conditions under which the electoral value of being seen to engage with relevant scientific expertise plausibly increased across political contexts. If legislators' baseline engagement with researchers is shaped in part by the audience costs of publicly associating with academic expertise, then a context that raises the perceived value of such associations should produce observable shifts in their digital behaviors—and those shifts should be concentrated on researchers whose expertise is directly relevant to the crisis at hand.

I employ the expected changes in salience of some types of expertise to investigate whether there were discernible shifts in the digital engagement behaviors of legislators with researchers in times of crisis when the stakes are high and the demand for evidence should be pronounced.

## Results

**Do legislators follow and engage with researchers online?** The general descriptive statistics offer insights into the interactions that characterize their relationships on Twitter.

The majority of legislators follow at least one researcher account from the list (about 86% of legislators). The median legislator was following 990 accounts of which 8 were academic researchers (see *SI Appendix, Table B1* for an overview of the distribution of these behaviors). However, following is concentrated amongst only a portion of accounts from the list (roughly 93% of the researchers are not in the legislators' Twitter 'radar').

The researcher list contains a number of power users with unusually strong social media followings. In total, the dataset contains around 20 accounts with more than 1M followers and 500 with more than 100K. The finding remains stable when trimming the top 1% of followed accounts (keeping only researchers with less than 16K followers). Around 84% of the legislators follow at least one account from the restricted list and the median legislator follows 6 users. Figure 2a presents an overview of the legislator-researcher Twitter events as a percentage of the total events for each behavior.

It is worth noting that the legislator accounts are highly active content-generators on Twitter. Approximately 60% of legislators have a daily posting rate, and a substantial 91% post on average once a week. In spite of the large volume of content production by legislators, only a small fraction can be linked to researchers. For instance, although about 73% of the legislators with retweets has ever retweeted a research producer. Behaviors such as men-

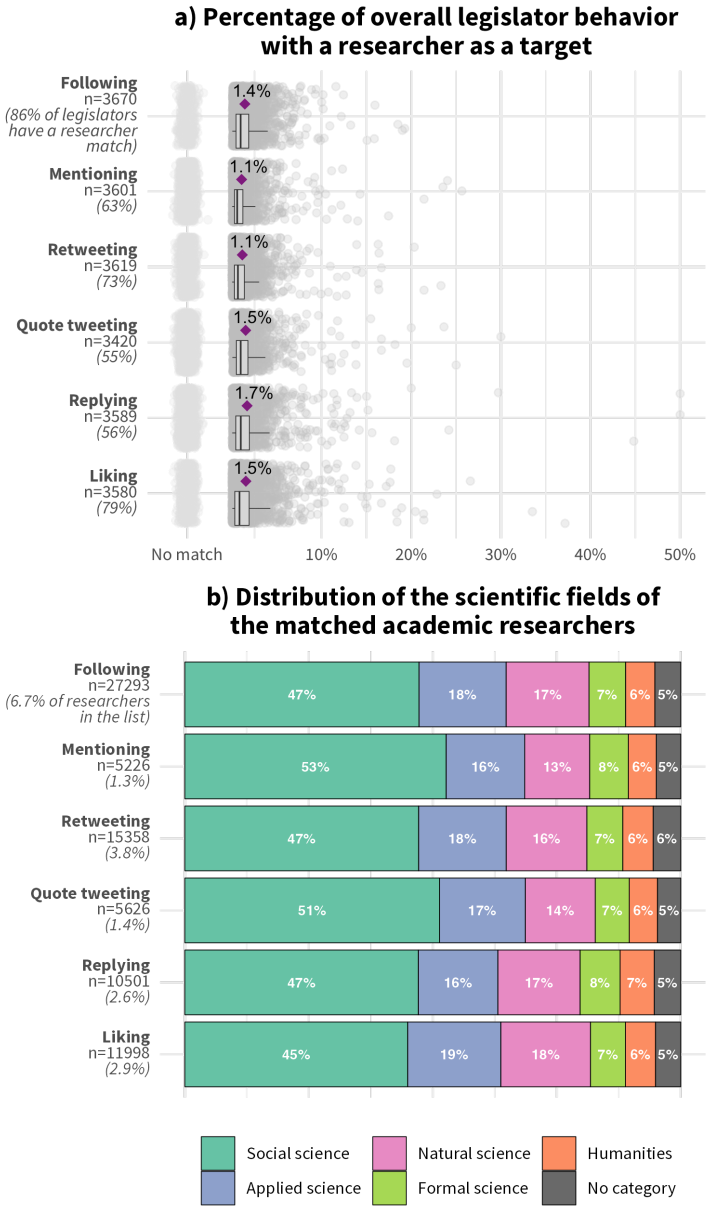


**Figure 2: Overview of legislator Twitter behaviors targeted at academic researchers.** In Panel A, dots represent individual legislators. The auxiliary information on the labels present the number of legislators who engage in the behavior and percentage who establish a researcher link at least once. Their position of the x-axis convey instances where events imply a legislator-researcher link as a proportion of the legislator totals. The purple rombi present the pooled legislator average. In Panel B, the auxiliary information denote the numbers and percentages of unique researchers from the original list identified as targets of legislators' Twitter events.

tioning an original posts, replying to, and quote tweeting are even rarer with slightly more than half of the legislators in the sample engaging in them and the median legislator doing it only once.

The data availability of legislator 'likes' is constrained by Twitter API limits, which allowed to retrieve latest 3,200 posts liked by a user. This means that these numbers present a lower bound, since the full 'like' history is unobserved for roughly 30% of legislators who presumably use the feature more often. Overall, 79% of legislators liked at least one post by an academic researcher.

Another set of interesting patterns comes from the researcher-end features. In Figure 2b, I present an overview of the proportional make up of the researchers legislators interact with by their scientific discipline. These data present a picture in which social scientists consistently account for about half of those researchers linked to legislators across behaviors. This is substantially larger than researchers in other fields. For instance, around 19% of listed political scientists are followed by at least one legislator, compared to 3.4% of listed biologists. The overall pool of researchers in the study operate in diverse branches, e.g., natural (32%), social (27%), and applied (24%) sciences. Still, social scientists in the list are about 3.5x more likely to be followed against a comparable group in absolute size, natural scientists, and account for almost 47% of all followed researchers in the pooled network.

All-in-all, these descriptive analyses suggest that legislators do follow and engage with researchers in the 'digital wild'. Still there is an important qualification to these findings. These instances are highly uncommon, accounting for slightly more than 1% of legislator Twitter events of each behavior. *SI Appendix, Figure B2* provides some evidence of this qualification by analyzing the structure of the following lists of other groups—specifically, political and science journalists, as well as a random sample of researchers. The networks of these users can serve as a benchmark as their professional incentives could also make them likely to follow and engage with researchers. The comparison highlights that while legislators do connect with researchers, they do so far less frequently than these other groups.

In addition to that, these behaviors are aimed at a select group of researchers. While 6.7% of researchers in the list of 410K appear in legislator networks, as behaviors become more active the group reduces between 1% and 4%. Furthermore, researchers in the social sciences account consistently for about half of the legislator-researcher matches across behaviors, suggesting that legislators tend to pay attention to them to a higher degree.

**Are there any predictors of engagement with researchers based on specific contextual and legislator characteristics?** Previous research has identified individual-level correlates of trust and politicization of science, as well as preferences for the role of scientific experts in policy debates (for an overview see Rutjens, Heine, et al., 2018). Overall, findings point to asymmetries related to political ideology (Gauchat, 2012; Lewandowsky and Oberauer, 2016; Linden et al., 2021; Funk et al., 2019), religiosity and spirituality (Rutjens, Sutton, and Lee, 2018; Rutjens, Sengupta, et al., 2022), knowledge about science and education level (Rutjens, Sutton, and Lee, 2018; Mede and Schäfer, 2020), age (Anderson et al., 2012), and gender (Gauchat, 2012; Roten, 2004). More recently, cross-national evidence of science skepticism in 24 countries suggests that there is variation of predictors across politicized scientific domains, in addition to heterogeneity in the levels of skepticism across countries (Rutjens, Sengupta, et al., 2022). The signaling account developed above predicts that these patterns should extend to legislators' publicly visible engagement with researchers: if engagement carries audience costs that vary with the ideological composition of legislators' electoral coalitions, then ideology should be among the strongest and most consistent predictors of online engagement with academic expertise.

Figure 3 presents the results from a linear mixed-effect model exploring predictors of the proportion of researchers in legislators' networks. The general direction of the predictors is parallel in all the behaviors of the study (see *SI Appendix, Figures C1-C5* for the models of the remaining engagement behaviors and *Table B10* for model output).

The findings are broadly consistent with an account under which the political costs of publicly associating with academic expertise can vary systematically across ideological contexts. The most striking pattern concerns party affiliation, a picture consistent with previous findings in the general public (Gauchat, 2012; Lewandowsky and Oberauer, 2016; Mede and Schäfer, 2020). While legislators from most party families exhibit broadly similar engagement rates, those from conservative and radical right parties are significantly less likely to follow and engage with academic researchers, and do so at a lower rate when they do engage. The contrast with green party legislators is stark: around 99% of legislators from green parties follow at least one researcher, compared to 66% of those from nationalist and radical right parties.

Among the legislators in the sample on Twitter, 9% hold, or were enrolled in programs, leading to an advanced research qualification, such as doctorates or equivalent. These legislators with research qualifications themselves were more likely to follow and engage with academic researchers. Additionally, academic researchers represent a higher proportion of these behaviors in the platform for legislators with research backgrounds compared to their non-researcher counterparts. The patterns attached to these legislators with higher educational qualifications can stem from their knowledge and interest about science, as well as the personal networks they created in the process of obtaining their degrees.

While there are some national-level differences as evidenced in Figure 1b, there are no discernible regional differences. Although legislators exhibit a similar propensity to follow

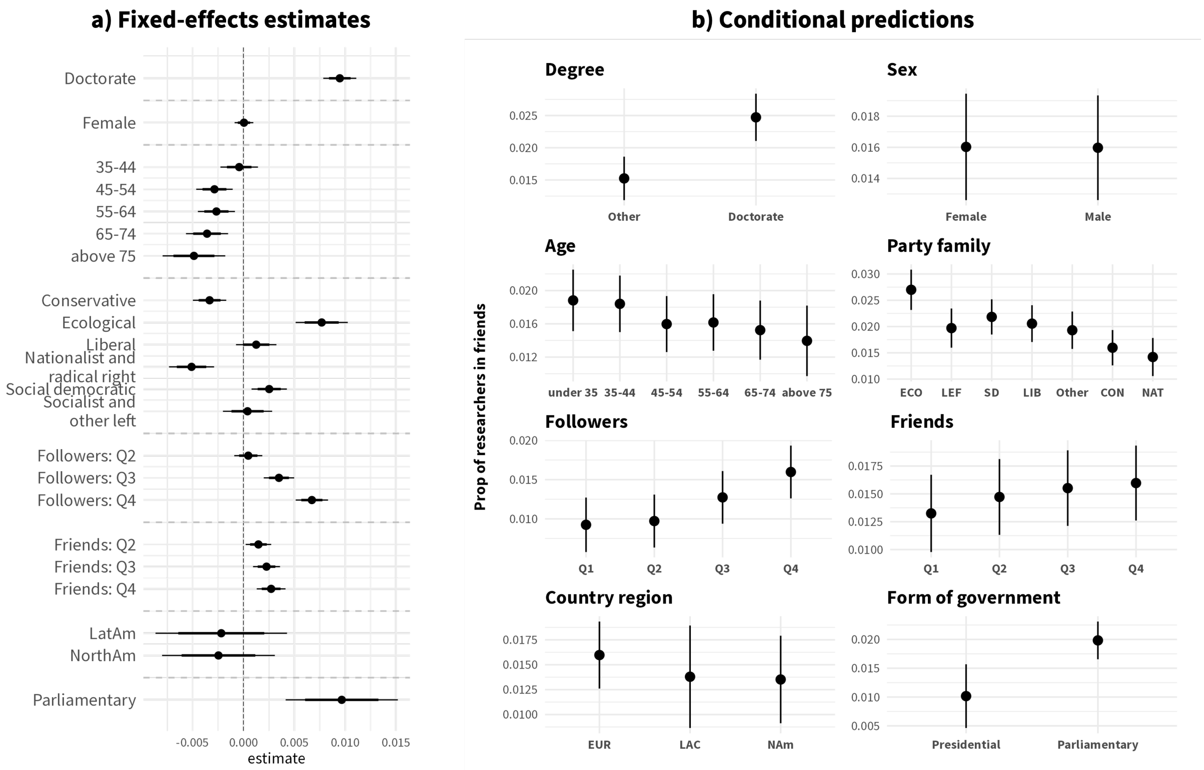


**Figure 3: Estimated effects of legislator and legislature characteristics on the proportion of researchers in their networks.** Results from a linear mixed-effects model with country random effects with age (under 35), party family (other), country region (Europe), system (presidential), and Q1 for followers and friends as references for categorical variables. Number of observations: 3,381; Groups: 12. Panel a presents the coefficients with 80% and 95% confidence intervals. The conditional predictions are computed with numeric covariates are held at their means and the other covariates at their modes: no research degree, presidential, European, male, 45-54, Q1, and Conservative party. The std. followers and friends are referenced to individual's legislature mean.

and engage with scientists in the list, the disparity becomes evident in the magnitude of their engagement. Latin American legislators, on average, follow and engage with a smaller absolute number of academic researchers compared to their counterparts from other regions. A potential explanation for the regional patterns can be on the researcher supply side and their proximity to the legislators (see *SI Appendix, Table B3* for an overview of the geographical distribution of the researchers).

Furthermore, researcher producers in the list consistently represent a higher proportion of the networks and behaviors of younger, "digital native" and more popular legislator accounts. Notably, there were no observable differences between male- and female-identifying legislators.

**Can legislators' inclination to follow and engage with researchers change? Evidence from a global pandemic.** The early stages of the COVID-19 pandemic, marked by unfamiliarity and uncertainty, provide a scenario to explore the implications of exogenous shocks to the demand of specific types of expertise on legislators' online behaviors and whether their publicly visible engagement with researchers is fixed or context-dependent.

A number of studies about the early reactions in politics to the pandemic have investigated shifts in public opinion formation and electoral outcomes (Bol et al., 2021; Leininger and Schaub, 2023), as well as changes in the content of politicians online communications (Kim et al., 2022; Guntuku et al., 2021; Engel-Rebitzer et al., 2021). In this part of the study, I focus on Twitter as a marketplace of information and assess the temporal variation of legislators' Twitter behaviors towards academic researchers.

These analyses rely on the rise of the COVID-19 pandemic as a source of variation in the importance of scientific expertise. Given the magnitude, unexpectedness and salience of the pandemic, I leverage this variation to assess the average observed changes in legislator behaviors between the pre- and during-COVID periods. I use January 30, 2020 as the cut-point for the periods. This date marks the declaration of COVID as a public health emergency of international concern by the World Health Organization, as well as coincides with increased public awareness of the virus and its risks (see *SI Appendix, Figure B3* for an overview of COVID-related search term popularity around the period).

The academic community swiftly reacted to the emergence of COVID-19. Surveys conducted among researchers in U.S. and European institutions reveal that approximately one-third of them shifted their focus to COVID-19-related research during the early phases of the pandemic across various disciplines (Gao et al., 2021). Even more, some research suggests that this scientific production quickly permeated policy circles, for example as source material for policy documents (Yin et al., 2021).

Figure 4 presents an overview of the distribution of the monthly legislator-researcher matches over the five-year period between January 2018 to December 2022. Across behaviors, the first half of 2020 is consistently among the periods with the highest number of instances in which legislators follow and engage with researchers during the time window. Most behaviors peak on March 2020, the month when COVID-19 was declared a pandemic.

Based on the literature on policy learning and diffusion indicating that experts with specialized knowledge can gain salience and wield influence in situations dealing with new and technically complex policy issues (Haas, 1992; Dunlop, 2017), I expect that at the dawn of the COVID pandemic, researchers with expertise pertaining to the crisis will see increases in engagement by legislators. To test this expectation, I look at the potential for changes in the rate of legislator-researcher links across Twitter behaviors between the pre- and during-COVID periods in the early stages of the pandemic.

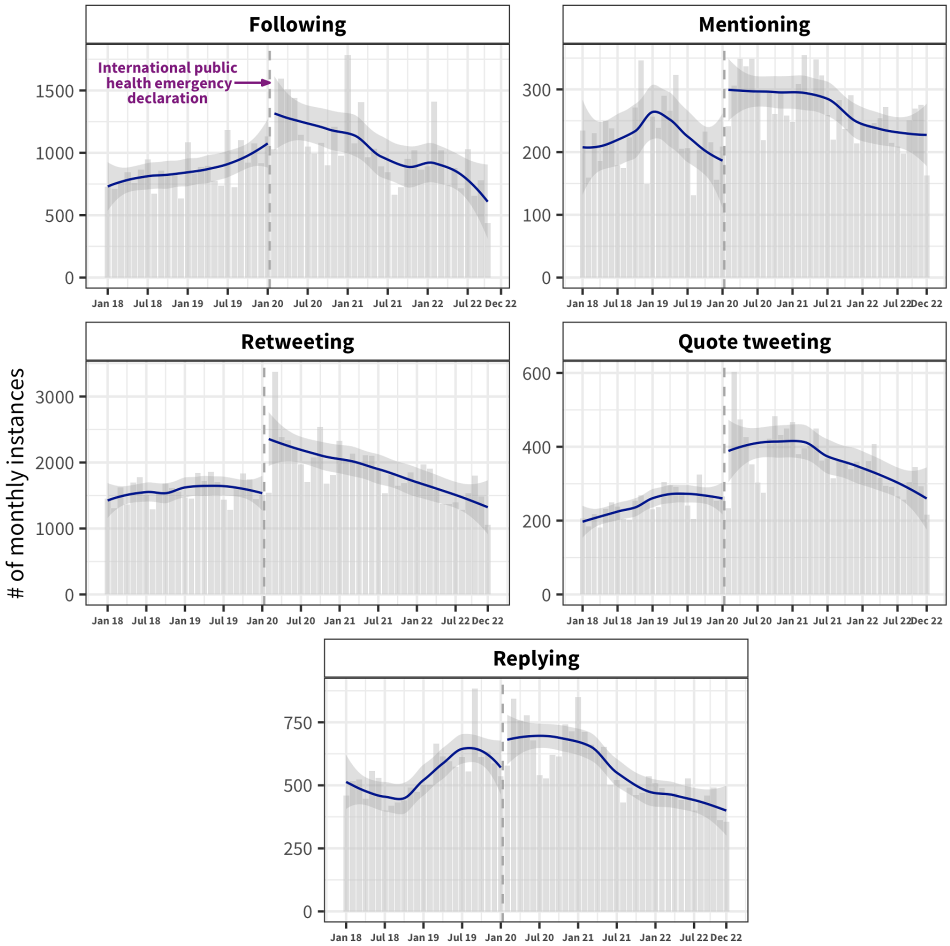


**Figure 4: Distribution of the counts of monthly legislator-researcher instances by behavior between 2018 and 2022 (with LOESS smoothed moving average).** The vertical dashed line marks the declaration of COVID-19 as an international public health emergency by the World Health Organization on January 30, 2023.

That said, there can be some potential dynamics at play that are not directly related to lawmakers' incentives, but reflect the broader impact that COVID could have had on science production and online behaviors more generally. For instance, potential changes in the supply of information or time spent online. It is possible that Twitter feeds contained more "scientific" content in the outset of COVID and people spent more time online. Still, it is not a logical consequence that politicians engage more when there is more science in their feed, rather one would expect that content needs to be relevant to translate into active behaviors.

Internet use increased during the pandemic (International Telecommunication Union, 2021). This seems to be also the case for legislators, who became more active on Twitter in the outset of the COVID period. There are observable increases across the range of platform-behaviors after January 2020, the only behavior that does not exhibit statistically significant changes is following ($B$ = 3.4; 95% CI=[-3.7,10.7]; $P$=0.34). The volume of following new accounts is constant over the pre- and post-threshold periods (see *SI Appendix, Figure C12* for

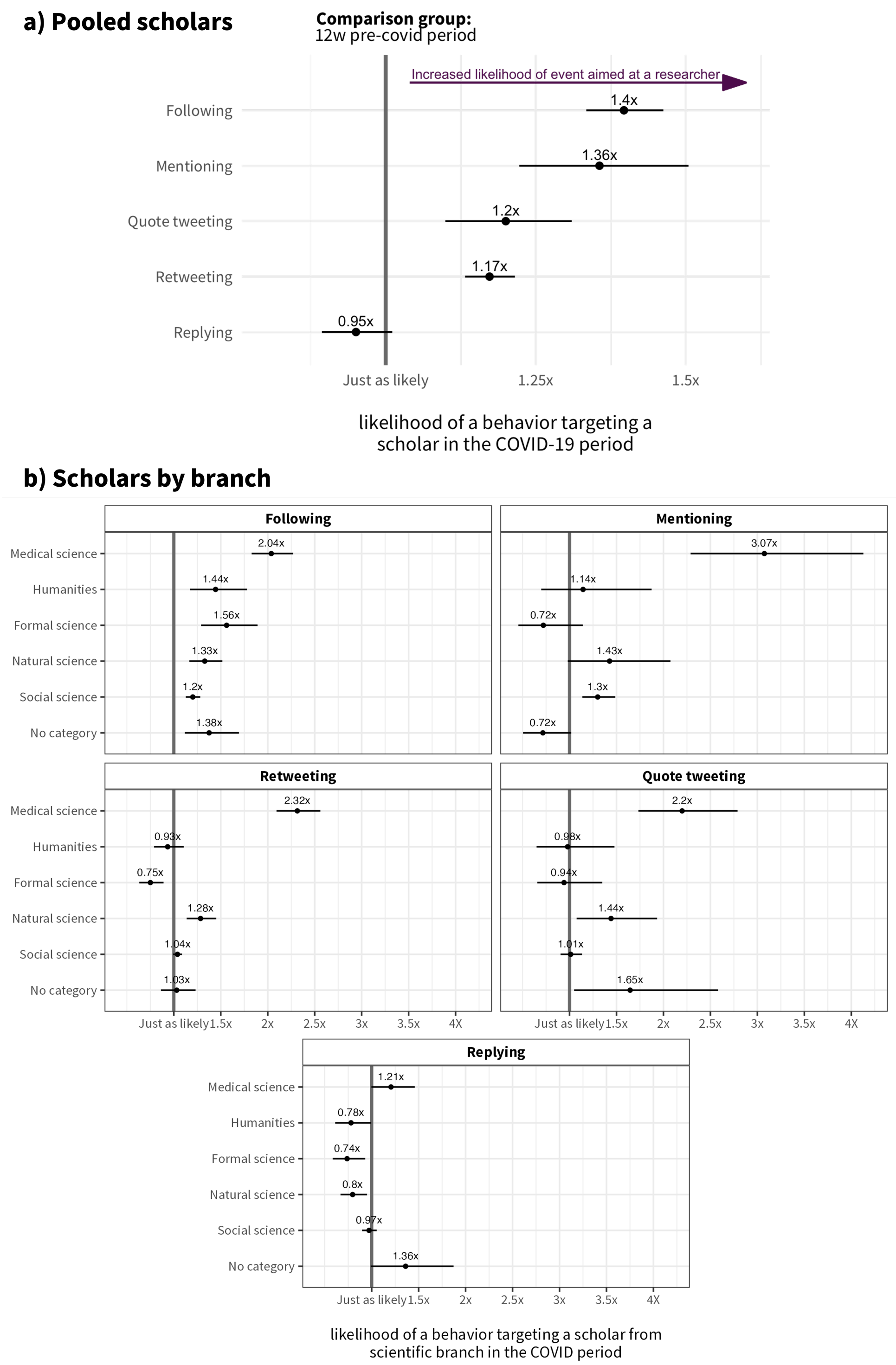


**Figure 5: Marginal effects on following and engagement with academic researchers during the COVID versus pre-COVID periods with a ±12 week bandwidth**. Results from a logistic mixed-effects models with country of legislature random effects. The estimates in the figure are relative risks representing the ratio of the probability of an event in the COVID period to the probability of an outcome in a pre-COVID period.

models exploring period differences across different bandwidths). That is to say, legislators tweeting and engagement with tweets increased on average in the during-COVID period, yet the rate at which they created new links in their network remained unchanged.

Since the increase in online activity at the outset of the pandemic can obfuscate the analysis of absolute changes between the two periods, I focus my analysis on the estimated differences in the ratio of the probability of an event in the COVID period to the probability of an outcome in an pre-COVID period.

To achieve this, I estimate logistic mixed-effects models with country random effects to explore the likelihood that an observed legislator Twitter event targets a researcher in the COVID compared to the pre-COVID period. Figure 5a presents the marginal effects extracted from the models comparing a ±12-week time-window since the declaration of COVID-19 as public health emergency of international concern.

The results suggest that researchers were more likely to be at the receiving end of legislator behaviors on Twitter in the first months of COVID compared to the baseline period. The only behavior that showcases no discernible change is a behavior that requires scholars to be first-movers by addressing legislators directly in their content in the first place, i.e., replying. For instance, new following edges in the 12 weeks after the declaration of COVID-19 as public health emergency of international concern period are 40% ($B$ = 1.4; 95% CI=[1.33,1.46]; $P$=<0.001); more likely to be targeted at a scholar compared to new edges created in the pre-COVID period.

These findings present evidence that legislators' inclination to follow and engage with researchers increased in the period where expertise from epistemic communities gained salience. Notably, according to subgroups analyses the increases are not observed for populist left and right legislators (see *SI Appendix, Figure C13 and C14* for the party family and education subgroup marginal effects).

Although previous evidence suggests that COVID research transcended disciplinary boundaries, there are signs that at the initial stages particular attention was paid to biomedical research (Yin et al., 2021). Considering legislators' online behaviors at the dawn of the pandemic, we could anticipate a similar tendency.

That is, if the changes in behaviors reflected the increased interest, need for expertise, and interactions with scholars, we would expect them to be particularly targeted at researchers whose expertise is more closely related to the immediate questions emanating from the rise of COVID.

The models in Figure 5b look at the different behaviors targeting scholars in the specific groups based on their scientific branch. The subgroup that sticks out the most are the researchers belonging to the medical sciences. While the increase in the likelihood of a new following event being targeted at scholars is observable across all fields, the other behaviors present a picture where researchers from the medical field experience the largest increases in the COVID period and in some cases the only ones experiencing increases compared to the baseline period. For example, these results suggest that retweets observed in the COVID period were 2.3 times more likely to have a medical scientist as a source creator compared to retweets in the pre-COVID period. Although expertise and commentary from researchers in other branches could have been of great importance during this period, arguably, researchers in this subgroup are the more plausible members of the epistemic communities holding specialized knowledge for this particular situation.

Altogether, the temporal analyses of legislator behaviors on Twitter reveal an increase in engagement with researchers. The observed surge in legislator-to-researcher interactions evident across behaviors underscores the importance of specialized knowledge in informing the public discussion at the dawn of the COVID pandemic. Notably, the growth in attention to scholars in the biomedical research camp reflects the relative importance of expertise relevant to the immediate challenges posed by the pandemic.

## Discussion

Idealized models often depict policymaking as a linear process of rational problem-solving, where decision-makers objectively integrate available evidence. In practice, lawmakers navigate a more complex reality. Their behavior is shaped by institutional, ideological, and electoral constraints that influence which information they seek, trust, and use (Cairney, 2016; Parkhurst, 2017; Hassel and Wegrich, 2022).

The growing paradigm of evidence-informed policymaking acknowledges these challenges while emphasizing the value of connecting scientific research with decision-making. However, research evidence is not always timely, salient, or packaged in ways that resonate with policymakers' agendas (Senninger and Seeberg, 2024; Walgrave and Dejaeghere, 2017). As researchers compete with other information sources for policymakers' attention, understanding how—and whether—lawmakers encounter academic expertise is increasingly relevant. Yet this question has a dimension that requires some attention: engagement with researchers is not only an informational act but also a public one, visible to the constituencies, party elites, and media audiences that legislators must attend to. Understanding when and how lawmakers publicly engage with researchers therefore requires attending not only to their informational needs but also to the political context that shapes the costs and benefits of being seen to do so.

This study explores one observable dimension of that relationship: legislators' publicly visible engagement with researchers on social media. While traditionally, these relationships may have been studied through more formal channels, such as committee meetings or research briefs, this study takes advantage of the new, more informal ways lawmakers and researchers can now connect. I structured my inquiry following three sequential questions regarding the prevalence, correlates, and malleability of legislator engagement with academic researchers online.

First, legislators do engage with researchers online, but this engagement is rare. In the pooled sample, 86% of legislators followed at least one researcher, and between 60 and 80% engaged in one of the platform-specific behaviors at least once. Yet the proportion of behaviors targeted at researchers relative to totals is roughly 1% across the board—meaning that on average, researchers constitute just 1% of legislators' networks and of the content they actively engage with. The median legislator has retweeted only 4 posts by researchers and quoted, mentioned, or replied to researchers just once in their entire Twitter history. Comparisons with science and political journalists—whose professional incentives also create reasons to follow researchers—underscore the gap: journalists follow researchers at substantially higher rates than legislators do. These findings suggest that while the infrastructure for legislator-researcher contact exists on social media, it might be activated only selectively and for a narrow set of researchers, with social scientists consistently accounting for a disproportionate share of the researchers legislators attend to

Second, engagement varies systematically across legislators in ways consistent with a signaling account. Legislators with doctoral degrees are more likely to follow and engage with researchers, and researchers constitute a larger share of their online networks—a pattern consistent with both genuine interest in academic knowledge and the personal networks formed during graduate training. Also theoretically salient is the ideological divide. Lawmakers from green parties are significantly more likely to follow and engage with researchers, and do so at higher rates proportionally. By contrast, legislators from conservative and radical right parties, ideological families associated with science skepticism and anti-expert populism, exhibit systematically lower engagement. This partisan asymmetry is consistent with the view that publicly visible engagement with academic expertise might carry different audience costs depending on a legislator's electoral coalition: for those representing constituencies broadly supportive of scientific institutions, engaging with researchers is relatively costless; for those whose coalitions mobilize skepticism toward expert elites, it entails greater political risk. This disparity is not merely a curiosity of online behavior—it has potential implications for lawmaking, with partisan divergence in willingness to engage with the academic community potentially shaping the extent to which evidence reaches the policy process (e.g., Furnas, LaPira, and Wang, 2025).

Third, the constraints on publicly visible engagement are context-dependent rather than fixed. The onset of the COVID-19 pandemic provides a critical test of this claim. If baseline engagement is shaped in part by audience costs, then a context that raises the perceived legitimacy and electoral value of engaging with expert knowledge should produce observable increases in legislators' publicly visible engagement with researchers. Also important, those increases should be targeted at researchers whose expertise is most directly relevant to the crisis. I find support for both expectations in the data. Across most behaviors, legislators were significantly more likely to target researchers in the COVID period relative to the pre-COVID baseline, even after accounting for the general increase in online activity. Crucially, these increases were concentrated on researchers in the medical sciences, who experienced the largest and in some cases the only statistically significant increases in engagement across behaviors. A retweet in the COVID period was 2.3 times more likely to feature a medical scientist than in the pre-COVID period. This pattern of targeted rather than diffuse engagement is difficult to explain as a byproduct of increased scientific content supply or general online activity—it points instead to legislators selectively amplifying the expertise most pertinent to the policy challenge at hand. Importantly, the increases are not observed for populist left and right legislators, suggesting that even under conditions of heightened epistemic demand, audience cost constraints do not disappear entirely for legislators whose political identities are bound up with anti-expert positioning.

Taken together, these findings offer a new perspective on the relationship between political elites and academic expertise. Legislators' publicly visible engagement with researchers is neither random nor uniform: it is likely shaped by the ideological composition of electoral coalitions, by individual characteristics such as academic background, and by contextual shifts in the salience and legitimacy of expertise. The logic developed in this study does not require that legislators consciously weigh each interaction as a strategic calculation—it simply suggests that the broader political environment structures the incentives within which engagement occurs, and that those incentives vary in ways that leave observable traces in digital behavioral data.

That said, some limitations to this approach deverve to be acknowledged. The behavioral measures I employ capture public-facing engagement but cannot speak to the content or quality of legislator-researcher interactions, nor to whether such engagement influences legislative behavior or policy outputs. The data reflect a single platform at a particular moment in its history, and the pre-2022 Twitter environment may not be representative of elite digital behavior on successor platforms. The researcher database, while large, captures only those academics who were active on Twitter and shared scholarly links—a sample that may skew toward researchers already oriented toward public engagement and science communication. These constraints mean that the findings speak to a visible margin of elite-expert contact rather than to the full range of ways lawmakers encounter and use research.

A further limitation concerns the platform itself. Twitter has changed substantially since the data were collected, and the pre-2022 environment studied here may have limited temporal validity as a window onto contemporary elite digital behavior (Munger, 2023). The platform's transformation after 2022—in ownership, moderation, and user composition—makes it unlikely that the behavioral patterns documented here would replicate in the current environment, and raises broader questions about the generalizability of findings from any single platform at a particular historical moment. At the same time, this limitation has a counterpart that is worth making explicit. The pre-2022 Twitter was a genuinely unusual observational setting: it attracted disproportionately high concentrations of both political elites and academic researchers, who operated within a shared public network in ways that made their interactions tractable at scale. The cohabitation of these two communities on a single platform—in a period when both groups were authentically present and behaviorally active—is unlikely to be replicated on successor platforms. The findings are therefore best understood as documenting a bounded but analytically valuable window of elite-expert contact in the digital public sphere, rather than as claims about online political behavior in general.

Within those bounds, the signals uncovered here speak to a crucial precondition in the chain of events that could lead to research permeating the halls of legislatures: whether legislators engage with researchers and the knowledge they produce in the first place. The findings suggest that this precondition is met unevenly, concentrated among legislators with particular ideological orientations and educational backgrounds, and most reliably activated when the political environment makes expertise salient. Understanding the conditions under which those barriers ease, and those under which they persist even in crisis, is an important task for future research on the politics of evidence and expertise.

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## Data availability

The data and scripts necessary to reproduce the results in this article are available at: —

## Competing interests

The author declares no competing interests.

## Ethical standards

This study received ethics approval by the — Research Ethics Office (Application ID—)

## References

Anderson, Ashley A., Dietram A. Scheufele, Dominique Brossard, and Elizabeth A. Corley (2012). “The Role of Media and Deference to Scientific Authority in Cultivating Trust in Sources of Information about Emerging Technologies”. *International Journal of Public Opinion Research* 24.2, pp. 225–237.

Avey, Paul C. and Michael C. Desch (2014). “What Do Policymakers Want From Us? Results of a Survey of Current and Former Senior National Security Decision Makers”. *International Studies Quarterly* 58.2, pp. 227–246.

Baekgaard, Martin, Julian Christensen, Casper Mondrup Dahlmann, Asbjørn Mathiasen, and Niels Bjørn Grund Petersen (2019). “The Role of Evidence in Politics: Motivated Reasoning and Persuasion among Politicians”. *British Journal of Political Science* 49.3, pp. 1117–1140.

Barberá, Pablo (2015). “Birds of the Same Feather Tweet Together: Bayesian Ideal Point Estimation Using Twitter Data”. *Political Analysis* 23.1, pp. 76–91.

Bavel, Jay J. Van, Katherine Baicker, Paulo S. Boggio, Valerio Capraro, Aleksandra Cichocka, Mina Cikara, Molly J. Crockett, Alia J. Crum, Karen M. Douglas, James N. Druckman, John Drury, Oeindrila Dube, Naomi Ellemers, Eli J. Finkel, James H. Fowler, Michele Gelfand, Shihui Han, S. Alexander Haslam, Jolanda Jetten, Shinobu Kitayama, Dean Mobbs, Lucy E. Napper, Dominic J. Packer, Gordon Pennycook, Ellen Peters, Richard E. Petty, David G. Rand, Stephen D. Reicher, Simone Schnall, Azim Shariff, Linda J. Skitka, Sandra Susan Smith, Cass R. Sunstein, Nassim Tabri, Joshua A. Tucker, Sander van der Linden, Paul van Lange, Kim A. Weeden, Michael J. A. Wohl, Jamil Zaki, Sean R. Zion, and Robb Willer (2020). “Using social and behavioural science to support COVID-19 pandemic response”. *Nature Human Behaviour* 4.5, pp. 460–471.

Berger, Loıc, Nicolas Berger, Valentina Bosetti, Itzhak Gilboa, Lars Peter Hansen, Christopher Jarvis, Massimo Marinacci, and Richard D. Smith (2021). “Rational policymaking during a pandemic”. *Proceedings of the National Academy of Sciences* 118.4, e2012704118.

Bogenschneider, Karen, Elizabeth Day, and Bret N. Bogenschneider (2021). “A window into youth and family policy: State policymaker views on polarization and research utilization”. *The American Psychologist* 76.7, pp. 1143–1158.

Bol, Damien, Marco Giani, André Blais, and Peter John Loewen (2021). “The effect of COVID-19 lockdowns on political support: Some good news for democracy?” *European Journal of Political Research* 60.2, pp. 497–505.

Bornmann, Lutz, Robin Haunschild, and Rüdiger Mutz (2021). “Growth rates of modern science: a latent piecewise growth curve approach to model publication numbers from established and new literature databases”. *Humanities and Social Sciences Communications* 8.1, pp. 1–15.

Brainard, Jeffrey (2022). “Riding the twitter wave”. *Science* 375.6587, pp. 1344–1347.

Bunea, Adriana, Raimondas Ibenskas, and Florian Weiler (2025). “Power, policy and national background: understanding interest groups-legislators ties on social media in the European Parliament”. *Journal of European Public Policy*, pp. 1–27.

Cairney, Paul (2016). *The Politics of Evidence-Based Policy Making*. Springer.

Caplan, Nathan (1979). “The Two-Communities Theory and Knowledge Utilization”. *American Behavioral Scientist* 22.3, pp. 459–470.

Caplan, Nathan, Russell J. Stambaugh, Andrea Morrison, and University of Michigan (1975). *The use of social science knowledge in policy decisions at the national level: a report to respondents*. Ann Arbor: Institute for Social Research, University of Michigan.

Castanho Silva, Bruno and Sven-Oliver Proksch (2022). “Politicians unleashed? Political communication on Twitter and in parliament in Western Europe”. *Political Science Research and Methods* 10.4, pp. 776–792.

Dodson, Elizabeth A., Nora A. Geary, and Ross C. Brownson (2015). “State legislators’ sources and use of information: bridging the gap between research and policy”. *Health Education Research* 30.6, pp. 840–848.

Dunbar, R. I. M., Valerio Arnaboldi, Marco Conti, and Andrea Passarella (2015). “The structure of online social networks mirrors those in the offline world”. *Social Networks* 43, pp. 39–47.

Dunlop, Claire A. (2017). “The irony of epistemic learning: epistemic communities, policy learning and the case of Europe’s hormones saga”. *Policy and Society* 36.2, pp. 215–232.

Engel-Rebitzer, Eden, Daniel Camargo Stokes, Alison Buttenheim, Jonathan Purtle, and Zachary F. Meisel (2021). “Changes in legislator vaccine-engagement on Twitter before and after the arrival of the COVID-19 pandemic”. *Human Vaccines & Immunotherapeutics* 17.9, pp. 2868–2872.

Fortunato, Santo, Carl T. Bergstrom, Katy Börner, James A. Evans, Dirk Helbing, Staša Milojević, Alexander M. Petersen, Filippo Radicchi, Roberta Sinatra, Brian Uzzi, Alessandro Vespignani, Ludo Waltman, Dashun Wang, and Albert-László Barabási (2018). "Science of science". *Science* 359.6379, eaao0185.

Funk, Cary, Meg Hefferon, Brian Kennedy, and Courtney Johnson (2019). "Trust and Mistrust in Americans' Views of Scientific Experts".

Furnas, Alexander C., Timothy M. LaPira, and Dashun Wang (2025). "Partisan disparities in the use of science in policy". *Science* 388.6745, pp. 362–367.

Gao, Jian, Yian Yin, Kyle R. Myers, Karim R. Lakhani, and Dashun Wang (2021). "Potentially long-lasting effects of the pandemic on scientists". *Nature Communications* 12.1, p. 6188.

Garg, Prashant and Thiemo Fetzer (2025). "Political expression of academics on Twitter". *Nature Human Behaviour*, pp. 1–18.

Gauchat, Gordon (2012). "Politicization of Science in the Public Sphere: A Study of Public Trust in the United States, 1974 to 2010". *American Sociological Review* 77.2, pp. 167–187.

Geddes, Marc (2021). "The webs of belief around 'evidence' in legislatures: The case of select committees in the UK House of Commons". *Public Administration* 99.1, pp. 40–54.

Guntuku, Sharath Chandra, Jonathan Purtle, Zachary F Meisel, Raina M Merchant, and Anish Agarwal (2021). "Partisan Differences in Twitter Language Among US Legislators During the COVID-19 Pandemic: Cross-sectional Study". *Journal of Medical Internet Research* 23.6, e27300.

Haas, Peter (1992). "Introduction: Epistemic Communities and International Policy Coordination". *International Organization* 46, pp. 1–35.

Hassel, Anke and Kai Wegrich (2022). *How to Do Public Policy*. Oxford University Press.

He, Yuanmo and Milena Tsvetkova (2023). "A Method for Estimating Individual Socioeconomic Status of Twitter Users". *Sociological Methods & Research*, p. 00491241231168665.

Heuer, Ruth, Peter Einaudi, and Kelly Kang (2023). *Research Doctorate Conferrals Rebound, Leading to Record Number of U.S. Doctorate Recipients in 2022*. Tech. rep. NSF 23-353. National Science Foundation.

Huberman, Michael (1994). "Research utilization: The state of the art". *Knowledge and Policy* 7.4, pp. 13–33.

International Telecommunication Union (2021). *Measuring digital development - Facts and figures 2021*. Geneva: International Telecommunication Union.

Jones, Bryan D. and Frank R. Baumgartner (2005). *The Politics of Attention: How Government Prioritizes Problems*. Chicago, IL: University of Chicago Press.

Jungherr, Andreas (2016). "Twitter use in election campaigns: A systematic literature review". *Journal of Information Technology & Politics* 13.1, pp. 72–91.

Kim, Taegyoon, Nitheesha Nakka, Ishita Gopal, Bruce A. Desmarais, Abigail Mancinelli, Jeffrey J. Harden, Hyein Ko, and Frederick J. Boehmke (2022). "Attention to the COVID-19 Pandemic on Twitter: Partisan Differences Among U.S. State Legislators". *Legislative Studies Quarterly* 47.4, pp. 1023–1041.

Klar, Samara, Yanna Krupnikov, John Barry Ryan, Kathleen Searles, and Yotam Shmargad (2020). "Using social media to promote academic research: Identifying the benefits of twitter for sharing academic work". *PLOS ONE* 15.4, e0229446.

Lamberton, Cait and Andrew T. Stephen (2016). "A Thematic Exploration of Digital, Social Media, and Mobile Marketing: Research Evolution from 2000 to 2015 and an Agenda for Future Inquiry". *Journal of Marketing* 80.6, pp. 146–172.

Lee, Nathan (2022). "Do Policy Makers Listen to Experts? Evidence from a National Survey of Local and State Policy Makers". *American Political Science Review* 116.2, pp. 677–688.

Leininger, Arndt and Max Schaub (2023). "Strategic Alignment in Times of Crisis: Voting at the Dawn of a Global Pandemic". *Political Behavior*.

Lewandowsky, Stephan and Klaus Oberauer (2016). "Motivated Rejection of Science". *Current Directions in Psychological Science* 25.4, pp. 217–222.

Linden, Sander van der, Costas Panagopoulos, Flávio Azevedo, and John T. Jost (2021). "The Paranoid Style in American Politics Revisited: An Ideological Asymmetry in Conspiratorial Thinking". *Political Psychology* 42.1, pp. 23–51.

Mede, Niels G., Viktoria Cologna, Sebastian Berger, John Besley, Cameron Brick, Marina Joubert, Edward Maibach, Sabina Mihelj, Naomi Oreskes, and Mike S. Schäfer (2024). *Public communication about science across 68 countries: Global evidence on how people get information and communicate about science-related matters*.

Mede, Niels G. and Mike S. Schäfer (2020). "Science-related populism: Conceptualizing populist demands toward science". *Public Understanding of Science* 29.5, pp. 473–491.

Metaxas, Panagiotis, Eni Mustafaraj, Kily Wong, Laura Zeng, Megan O'Keefe, and Samantha Finn (2015). "What Do Retweets Indicate? Results from User Survey and Meta-Review of Research". *Proceedings of the International AAAI Conference on Web and Social Media* 9.1, pp. 658–661.

Mongeon, Philippe, Timothy D. Bowman, and Rodrigo Costas (2023). "An open data set of scholars on Twitter". *Quantitative Science Studies*, pp. 1–11.

Munger, Kevin (2023). "Temporal validity as meta-science". *Research & Politics* 10.3, p. 20531680231187271.

National Research Council (2012). *Using Science as Evidence in Public Policy*. Washington, D.C.: National Academies Press.

Ouimet, Mathieu, Morgane Beaumier, Adrien Cloutier, Alexandre Côté, Éric Montigny, François Gélineau, Steve Jacob, and Stéphane Ratté (2023). "Use of research evidence in legislatures: a systematic review". *Evidence & Policy* -1.aop, pp. 1–18.

Parkhurst, Justin (2017). *The Politics of Evidence: From evidence-based policy to the good governance of evidence*. Taylor & Francis.

Priem, Jason, Heather Piwowar, and Richard Orr (2022). *OpenAlex: A fully-open index of scholarly works, authors, venues, institutions, and concepts*.

Purtle, Jonathan, Elizabeth A. Dodson, Katherine Nelson, Zachary F. Meisel, and Ross C. Brownson (2018). "Legislators' Sources of Behavioral Health Research and Preferences for Dissemination: Variations by Political Party". *Psychiatric Services* 69.10, pp. 1105–1108.

Riphahn, Regina T. and Monika Schnitzer (2022). "Die Bedeutung wissenschaftlicher Evidenz für die Arbeit des Bundestags". *Wirtschaftsdienst* 102.4, pp. 270–272.

Roten, Fabienne Crettaz von (2004). "Gender Differences in Attitudes toward Science in Switzerland". *Public Understanding of Science* 13.2, pp. 191–199.

Rutjens, Bastiaan T., Steven J. Heine, Robbie M. Sutton, and Frenk Van Harreveld (2018). "Attitudes Towards Science". In: *Advances in Experimental Social Psychology*. Vol. 57. Elsevier, pp. 125–165.

Rutjens, Bastiaan T., Nikhil Sengupta, Romy van der Lee, Guido M. van Koningsbruggen, Jason P. Martens, André Rabelo, and Robbie M. Sutton (2022). "Science Skepticism Across 24 Countries". *Social Psychological and Personality Science* 13.1, pp. 102–117.

Rutjens, Bastiaan T., Robbie M. Sutton, and Romy van der Lee (2018). "Not All Skepticism Is Equal: Exploring the Ideological Antecedents of Science Acceptance and Rejection". *Personality and Social Psychology Bulletin* 44.3, pp. 384–405.

Seidel, Katja, Hans Verbeek, Sophie Fessel, and Florian Meer (2021). *Nutzen von wissenschaftlicher Evidenz: Erwartungen an wissenschaftliche Expertise : Bericht über eine Bundestagsbefragung im Rahmen der Leopoldina-Evidenzinitiative unter Leitung von Claudia Buch, Regina T. Riphahn und Monika Schnitzer*. Tech. rep. MyCoRe Community, vol. 27.

Senninger, Roman and Henrik B. Seeberg (2024). "Which Information Do Politicians Pay Attention To? Evidence from a Field Experiment and Interviews". *British Journal of Political Science*, pp. 1–18.

Spierings, Niels, Kristof Jacobs, and Nik Linders (2019). "Keeping an Eye on the People: Who Has Access to MPs on Twitter?" *Social Science Computer Review* 37.2, pp. 160–177.

Vivalt, Eva and Aidan Coville (2023). "How do policymakers update their beliefs?" *Journal of Development Economics* 165, p. 103121.

Walgrave, Stefaan and Yves Dejaeghere (2017). "Surviving Information Overload: How Elite Politicians Select Information". *Governance* 30.2, pp. 229–244.

Weiss, Carol H. (1979). "The Many Meanings of Research Utilization". *Public Administration Review* 39.5, pp. 426–431.

Weiss, Carol H. and Michael J. Bucuvalas (1980). "Truth Tests and Utility Tests: Decision-Makers' Frames of Reference for Social Science Research". *American Sociological Review* 45.2, pp. 302–313.

Westerman, David, Patric R. Spence, and Brandon Van Der Heide (2014). "Social Media as Information Source: Recency of Updates and Credibility of Information". *Journal of Computer-Mediated Communication* 19.2, pp. 171–183.

Wojcieszak, Magdalena, Andreu Casas, Xudong Yu, Jonathan Nagler, and Joshua A. Tucker (2022). "Most users do not follow political elites on Twitter; those who do show overwhelming preferences for ideological congruity". *Science Advances* 8.39, eabn9418.

Yin, Yian, Jian Gao, Benjamin F Jones, and Dashun Wang (2021). "Coevolution of policy and science during the pandemic". *Science* 371.6525, pp. 128–130.

# Politicians from 12 countries rarely engage with researchers on social media, but this can change when expertise gains salience

**Online Appendix**

## Contents

# Appendix A Data Collection, Sample, and Statistical Analyses

## A.1 Data on legislators from 12 countries

The data were collected between November and December 2022. The legislator database centers around the Twitter profiles of elected legislators during 2022. The data collected extends across 12 countries with widespread Twitter usage. The countries in the study include the three largest democracies in both South and North America (Argentina, Brazil, Colombia, Mexico, Canada, and the U.S.), in addition to six European democracies (France, Italy, Ireland, Germany, Spain, and the U.K.)

The data collection strategy is based on the Twitter IDs and handles extracted from three sources: legislators' (i) Wikidata entries (entities P6552 for Twitter IDs and P2002 for user names), (ii) Google Knowledge Graph, and (iii) official legislatures' webpages. In cases in which the initial Twitter ID extraction did not result in matches for a legislator, I performed a manual searches on Twitter under the individual's name. In a final step, I validated the accounts retrieved to ensure that they belonged to the legislator.

The legislators on Twitter constitute 90% of all legislators across the 13 legislatures, ranging from 72% in the Italian Chamber of Deputies to 100% in the US Senate.

Using all retrieved account IDs, I collected legislators' Twitter data using their API via the rtweet package (Kearney, 2019a) between November and December 2022. I extracted the legislators':

- **Account core:** A general overview of their accounts, including the date of creation, username, location, number of followers, amongst others.
- **Following network:** A snapshot of all the accounts they followed on November 30, 2022.
- **Tweets:** The universe of their public tweets, retweets, and quotes since they joined the platform until December 5, 2022.
- **Likes:** The archive of liked posts by each account with an upper bound of the 3,200 most recent.

In addition to collecting legislator's relevant Twitter data, I extracted auxiliary legislator sociodemographic and political information. In a first step, I mimicked the procedure by the Comparative Legislators Database (CLD) (Göbel and Munzert, 2022) to extract relevant information using Wikidata and Wikipedia as a data source. These data include legislators' political party affiliation from the legislatures' Wikipedia lists and their dates of birth (Wikidata entity P569). I use these variables in the predictive models. In a further step, I identify whether legislators have research credentials by having been enrolled or completed second-stage tertiary education degrees (e.g., doctorates or equivalent). I gathered this information from the official legislature's websites, government open repositories of public servant curricula, legislative transparency portals managed by academic institutions (e.g., Congreso Visible by the Universidad de los Andes in Colombia), legislators' personal website biographies, and Wikipedia entries.

## A.2 Data on researcher's social media profiles

The starting point for these data come from an open dataset of researchers on Twitter (Mongeon, Bowman, and Costas, 2023). The data builds on the Crossref Event Data release from January 2022 containing over 60 million Twitter posts with URL links to scholarly work. The authors extract the names of the authors associated to the works linked on the social media posts through the open repository of scholarly work, OpenAlex (Priem, Piwowar, and Orr, 2022). The connection between Twitter accounts to academic researchers relies on mapping the Twitter screen and user names onto the names of the authors associated to the papers mentioned in the posts. The authors outline a hierarchical matching process based on the name strings and variations of the handles and profile names. For instance, in one step matches are created, but not exclusively, after a full match between the authors name and the Twitter handle or user name (e.g., a paper whose author is Gregor Samsa is mentioned in a tweet by @gregorsamsa). In this dataset, Mongeon, Bowman, and Costas (2023) identify 423,920 Twitter accounts associated to authors of academic research studies with high precision.

Using these Twitter IDs, I collected researchers' core Twitter data using the Twitter REST API via the rtweet package (Kearney, 2019a) on December 12, 2022. When extracting the core features of these accounts, a total of 14,728 Twitter IDs did not relate to any active accounts.

Furthermore, I employed the OpenAlex author identifiers to extract auxiliary information about the individual researchers[1]. These data were retrieved as Author object calls from the OpenAlex API. The call returned last known institutional affiliations, total number of papers citing the author, and a list of concepts (inferred scientific fields) most frequently applied to research created by the author. Figure B1 and Table B3 present a descriptive overviews of the scientific disciplines and geographical associations of the identified researchers. The final researcher dataset contains core the Twitter and OpenAlex features of 409,192 identified authors of academic work.

Table B2 provides an overview of a selection of summary statistics from the legislators included in the study. Overall, the database contains Twitter, as well as sociodemographic and political information from 3,670 from a total set of 4,134 seating legislators (≈89%).

## A.3 Measurement

**Platform behaviors:** I map the list of research producer accounts onto the legislators' friendship snapshots, tweets, retweets, quotes, and likes. I encode instances where there is a legislator-academic researcher account pairing. For the analyses, I employ the binary measures of matches, total counts of instances, and proportions of matches in reference to totals for all platform-specific behaviors. In other words, whether the legislators follow, mention, retweet, quote, and like academic researchers, how many times they do, and what share that represents of the overall historic legislator behavior.

**Minimum possible following date:** An important set of data points to understand the temporal development of legislator Twitter networks are the dates in which they start to follow other accounts. Nevertheless, the platform does not make these data available through their API. One way to gather this information is to extract snapshots of the networks in different periods and comparing the relationships between nodes. Still, given the API limits, these crawls can be time expensive and only allow for tracking forward-looking developments. That is to say, they can only showcase changes from the first snapshot. In this study, I use a different strategy relying on the default call to extract users' friendships returning a reverse chronologically ordered list. Using these data, I employ a method for inferring the creation time of connections between users in unidirectional networks using auxiliary information from the nodes (Meeder et al., 2011). The procedure involves mapping legislators' chronologically ordered edge list alongside the "record-breakers" (users with the latest account creation times) for each legislator. The follow time for a user is estimated to

[1] These data were collected through between December 15-23, 2022, corresponding to the December 2022 snapshot of OpenAlex

be the creation time of the most recent record-breaker among the users who that legislator follows.

**Legislators' formal academic research background:** I collect the highest educational qualification of legislators. I extract this information from the i) legislature's websites, b) governmental repositories of public servant curricula, iii) legislative transparency portals managed by academic institutions, iv) legislators' personal sites, and v) Wikidata entries. I map legislators' qualifications on the International Standard Classification of Education (ISCED 97) (Hoffmeyer-Zlotnik and Wolf, 2003). I identify a legislator to have research credentials when they have been enrolled or completed second-stage tertiary education degrees (e.g., doctorates or equivalent).

**Scientific fields of academic researchers:** I employ the concepts attached to the work of academic researchers from OpenAlex to extract their predicted scientific field. This measure is based on a hierarchical representation of scientific concepts. OpenAlex uses a slightly modified concept tree based on the one developed by Microsoft Academic Graph (Shen, Ma, and Wang, 2018). I use the top layer of the tree containing 19 concepts ranging from political to materials science. These classes are derived from the titles, abstracts, and hosts of the papers authored by the researcher. Further, I map the predicted fields into four branches i) humanities, ii) social, iii) natural, and iv) formal science (see *SI Appendix, Figure B1* for an overview of the distribution of research fields).

**Party family:** I utilize the party family encoding from the Manifesto Project. In the main analyses, I group electoral alliances, single issue, and ethnic parties onto the other category, as well as christian-democratic and conservative parties onto an umbrella conservative party category. In the cases where parties were not present in the dataset, I assigned them to a category employing the Manifesto Party Family Handbook (Lehmann et al., 2023). *SI Appendix, Table B9* presents the different political parties alongside their family classification.

**Other covariates:** For the analyses, I employ: Age divided in six categories (under 35, 35-44, 45-54, 55-64, and above 75), sex measured using a dummy variable (male=0; female=1), and the accounts' following and friends quartiles.

## A.4 Statistical analyses

### A.4.1 Models on predictors of engagement with researchers based on contextual and legislator characteristics

To explore predictors of engagement with researchers, I ran linear mixed-effects models with country of legislature random effects. I specified the models with the lmer() function from the lme4 package in R (Bates, Mächler, et al., 2015). I report both fixed-effects estimates, as well as conditional predictions. I compute the conditional predictions with age (under 35), party family (other), country region (Europe), system (presidential), and Q1 for followers and friends as references for categorical variables using the marginaleffects package in R (Arel-Bundock, 2023).

I specified models for a) the proportions of researchers-legislator instances in respect to totals for each behavior (see main body and section C.1) and b) the absolute number of researchers-legislator instances (see section C.2). The linear mixed-effect model is given by:

$$y_{ij} = \beta_0 + \beta_1 x_{1ij} + \beta_2 x_{2ij} + ... + \beta_8 x_{8ij} + u_j + \epsilon_{ij}$$

where:

- $y_{ij}$ is the proportion or absolute number of scholars for observation $i$ in group $j$.
- $\beta_1, ..., \beta_8$ are the fixed effects coeffients for the covatiates.
- $u_j \sim N(0, \sigma_u^2)$ is the random intercept for each country $j$
- $\epsilon_{ij} \sim N(0, \sigma^2)$ is the residual error

### A.4.2 Models on changes in the likelihood that an observed legislator Twitter event targets a researcher in the COVID compared to the pre-COVID period

To explore potential changes in the composition of the targets of Twitter events between the pre- and during-COVID periods, I estimated logistic mixed-effects models with country random effects. I specified the models with the gmler() function from the lme4 package in R (Bates, Mächler, et al., 2015). I compute the relative risks representing the ratio of the probability of an event in the COVID period to the probability of an outcome in a pre-COVID period from the log-odds using the avg_comparisons() function from the marginaleffects package in R (Arel-Bundock, 2023).

I specified models for behaviors targeting scholars in the list (pooled), as well as by scientific branch. The generalized linear mixed-effects model is given by:

$$\text{logit}(P(Y_i = 1)) = \beta_0 + \beta_1 X_{i1} + u_j$$

where:

- $\text{logit}(P(Y_i = 1))$ is the log-odds of individual $i$ being a researcher or member of a specific research branch,
- $\beta_0$ is the fixed intercept,
- $\beta_1$ is the coefficient for the during COVID period $X_{i1}$ (Declaration of COVID as an international public health emergency),
- $u_j \sim \mathcal{N}(0, \sigma_u^2)$ is the random intercept for the country of legislation, with variance $\sigma_u^2$.

# Appendix B Supporting Figures and Tables

**Table B1: Distribution of total and researcher-targeted legislator behaviors**

| Behavior | Overall | | | | | | With academic researchers | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | No. of legs | Mean | Median | SD | Max | Min | No. of Legs[1] | Mean | Median | SD | Max | Min |
| Following | 3670 | 1766.5 | 990.0 | 3865.2 | 127189 | 1 | 3151 (85.9%) | 29.1 | 10 | 56.7 | 1146 | 1 |
| Mentioning | 3601 | 1102.4 | 562.0 | 1588.6 | 21726 | 1 | 2272 (63.1%) | 13.2 | 5 | 34.3 | 800 | 1 |
| Retweeting | 3619 | 3885.3 | 1476.0 | 8048.5 | 200781 | 1 | 2658 (73.4%) | 62.2 | 11 | 211.3 | 4901 | 1 |
| Quote tweeting | 3420 | 557.9 | 239.0 | 1036.9 | 23402 | 1 | 1880 (55%) | 12.9 | 4 | 36.4 | 940 | 1 |
| Replying | 3589 | 1317.6 | 370.0 | 3318.0 | 65903 | 1 | 2017 (56.2%) | 36.0 | 6 | 203.6 | 7980 | 1 |
| Liking | 3580 | 1808.7 | 1823.5 | 1254.5 | 3262 | 1 | 2828 (79%) | 31.7 | 12 | 55.4 | 726 | 1 |

[1] The percentage represents the share of legislators in relation to the number that engage in such behavior. For instance, 86% of the 3670 legislators that follow accounts on Twitter have an academic researcher match.

**Table B2:** Summary statistics of legislators included in the study

| | | | | Party family | | | | | | | Twitter Friends | | | Twitter Followers | | | Age | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | No. Legs | Research degree | Female | CON | ECO | LIB | NAT | SD | LEF | Other | Min | Max | Avg | Min | Max | Avg | Min | Max | Avg |
| Argentina | 61 | 3 (4.9%) | 26 (42.6%) | 7 (11.5%) | 0 (0%) | 0 (0%) | 0 (0%) | 17 (27.9%) | 0 (0%) | 37 (60.7%) | 12 | 7276 | 851.1 | 428 | 1177991 | 42331.1 | 34 | 79 | 56.2 |
| Brazil | 76 | 5 (6.6%) | 10 (13.2%) | 13 (17.1%) | 1 (1.3%) | 10 (13.2%) | 15 (19.7%) | 12 (15.8%) | 9 (11.8%) | 16 (21.1%) | 1 | 19936 | 1487.5 | 247 | 4025807 | 314161.7 | 39 | 80 | 59.0 |
| Canada | 324 | 14 (4.3%) | 101 (31.2%) | 111 (34.3%) | 2 (0.6%) | 155 (47.8%) | 0 (0%) | 25 (7.7%) | 0 (0%) | 31 (9.6%) | 1 | 26036 | 1739.1 | 181 | 6357427 | 39611.9 | 24 | 81 | 52.6 |
| Colombia | 104 | 3 (2.9%) | 32 (30.8%) | 27 (26%) | 9 (8.7%) | 23 (22.1%) | 4 (3.8%) | 0 (0%) | 19 (18.3%) | 22 (21.2%) | 27 | 24290 | 1890.4 | 168 | 1780086 | 113715.2 | 31 | 76 | 51.2 |
| France | 548 | 29 (5.3%) | 205 (37.4%) | 84 (15.3%) | 23 (4.2%) | 209 (38.1%) | 88 (16.1%) | 30 (5.5%) | 91 (16.6%) | 23 (4.2%) | 1 | 9676 | 1359.1 | 0 | 2894718 | 21771.4 | 22 | 79 | 48.8 |
| Germany | 583 | 95 (16.3%) | 207 (35.5%) | 127 (21.8%) | 110 (18.9%) | 90 (15.4%) | 64 (11%) | 153 (26.2%) | 35 (6%) | 4 (0.7%) | 1 | 8425 | 984.7 | 5 | 1072411 | 19399.4 | 24 | 74 | 47.3 |
| Ireland | 154 | 4 (2.6%) | 35 (22.7%) | 73 (47.4%) | 12 (7.8%) | 0 (0%) | 0 (0%) | 12 (7.8%) | 5 (3.2%) | 52 (33.8%) | 40 | 7493 | 1970.7 | 528 | 449917 | 20152.3 | 25 | 77 | 51.1 |
| Italy | 286 | 18 (6.3%) | 85 (29.7%) | 94 (32.9%) | 11 (3.8%) | 0 (0%) | 52 (18.2%) | 62 (21.7%) | 0 (0%) | 67 (23.4%) | 2 | 14081 | 993.5 | 0 | 1754299 | 36143.1 | 25 | 76 | 48.8 |
| Mexico | 114 | 20 (17.5%) | 61 (53.5%) | 21 (18.4%) | 5 (4.4%) | 0 (0%) | 0 (0%) | 26 (22.8%) | 59 (51.8%) | 3 (2.6%) | 12 | 39006 | 2242.8 | 237 | 3212451 | 117791.6 | 32 | 92 | 55.0 |
| Spain | 314 | 40 (12.7%) | 133 (42.4%) | 79 (25.2%) | 0 (0%) | 10 (3.2%) | 44 (14%) | 106 (33.8%) | 2 (0.6%) | 73 (23.2%) | 1 | 98546 | 2452.5 | 66 | 1717578 | 48252.2 | 26 | 75 | 51.9 |
| UK | 580 | 24 (4.1%) | 209 (36%) | 305 (52.6%) | 1 (0.2%) | 14 (2.4%) | 0 (0%) | 196 (33.8%) | 7 (1.2%) | 57 (9.8%) | 1 | 26769 | 2356.4 | 2 | 4776780 | 64919.6 | 26 | 82 | 52.0 |
| US House | 426 | 27 (6.3%) | 121 (28.4%) | 208 (48.8%) | 0 (0%) | 0 (0%) | 0 (0%) | 218 (51.2%) | 0 (0%) | 0 (0%) | 1 | 47205 | 1983.3 | 316 | 13464680 | 168925.0 | 27 | 86 | 59.1 |
| US Senate | 100 | 5 (5%) | 24 (24%) | 50 (50%) | 0 (0%) | 0 (0%) | 0 (0%) | 48 (48%) | 0 (0%) | 2 (2%) | 1 | 127189 | 4142.8 | 6838 | 12510878 | 755868.3 | 35 | 89 | 65.2 |
| **Pooled** | 3670 | 287 (7.8%) | 1249 (34%) | 1199 (32.7%) | 174 (4.7%) | 511 (13.9%) | 267 (7.3%) | 905 (24.7%) | 227 (6.2%) | 387 (10.5%) | 1 | 127189 | 1766.5 | 0 | 13464680 | 82174.9 | 22 | 92 | 52.0 |

## Figure B1: Overview of academic producers in the (Mongeon, Bowman, and Costas, 2023) list

| Predicted field | Twitter accounts | Followed by LEGs |
|---|---|---|
| **Pooled** | **409192** | 27295 (6.7%) |
| **Political science** (Social science) | 39172 | 7418 (18.9%) |
| **Economics** (Social science) | 8389 | 1404 (16.7%) |
| **Art** (Humanities) | 2054 | 317 (15.4%) |
| **History** (Humanities) | 3373 | 475 (14.1%) |
| **Philosophy** (Humanities) | 6179 | 820 (13.3%) |
| **Sociology** (Social science) | 4292 | 541 (12.6%) |

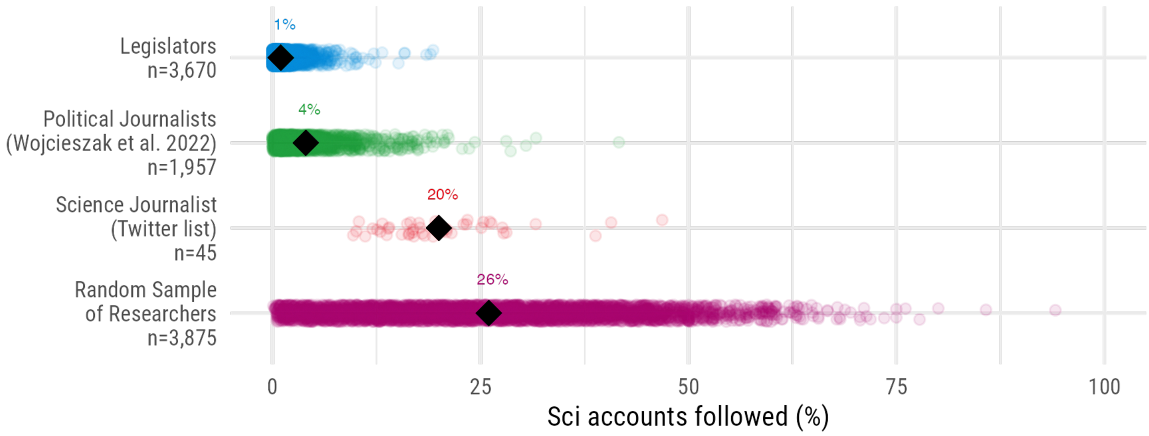
| Predicted field | Twitter accounts | Followed by LEGs |
|---|---|---|
| **Business** (Applied science) | 3690 | 362 (9.8%) |
| **Geography** (Natural science) | 4241 | 276 (6.5%) |
| **Psychology** (Social science) | 56815 | 3524 (6.2%) |
| **Computer science** (Formal science) | 29013 | 1527 (5.3%) |
| **Environmental science** (Natural science) | 6493 | 345 (5.3%) |
| **Geology** (Natural science) | 5005 | 257 (5.1%) |
| **Engineering** (Formal science) | 3346 | 167 (5%) |

| Predicted field | Twitter accounts | Followed by LEGs |
|---|---|---|
| **Medicine** (Applied science) | 94718 | 4439 (4.7%) |
| **Mathematics** (Formal science) | 7482 | 325 (4.3%) |
| **Physics** (Natural science) | 15217 | 555 (3.6%) |
| **Biology** (Natural science) | 83120 | 2814 (3.4%) |
| **Chemistry** (Natural science) | 12176 | 255 (2.1%) |
| **Materials science** (Natural science) | 2759 | 56 (2%) |
| **Not categorized** (No category) | 21658 | 1418 (6.5%) |

### Table B3: Geographical distribution of researchers based on their last known institutional affiliation

| Country | No. of acc | Country | No. of acc | Country | No. of acc | Country | No. of acc | Country | No. of acc |
|---|---|---|---|---|---|---|---|---|---|
| **United States** | 115101 | Indonesia | 621 | Costa Rica | 99 | Namibia | 20 | Nicaragua | 7 |
| **United Kingdom** | 59562 | Egypt | 608 | Iraq | 97 | Burundi | 19 | Uzbekistan | 7 |
| No info | 42970 | Nigeria | 585 | Tunisia | 96 | Mali | 19 | Angola | 6 |
| Australia | 19990 | Bangladesh | 529 | Lithuania | 95 | Albania | 18 | Gibraltar | 6 |
| **Canada** | 18589 | Ecuador | 508 | Kuwait | 93 | Congo - Brazzaville | 18 | Kyrgyzstan | 6 |
| **Spain** | 15604 | Malaysia | 504 | Bulgaria | 92 | Mongolia | 18 | Liechtenstein | 6 |
| **Germany** | 15362 | Taiwan | 504 | Malawi | 89 | Yemen | 18 | Papua New Guinea | 6 |
| **France** | 8825 | Philippines | 500 | Oman | 87 | St. Kitts & Nevis | 17 | Somalia | 6 |
| Netherlands | 8614 | Kenya | 496 | North Macedonia | 79 | Burkina Faso | 16 | Tajikistan | 6 |
| India | 8490 | Lebanon | 425 | Rwanda | 76 | Réunion | 16 | Eswatini | 5 |
| **Italy** | 8375 | Peru | 383 | Ukraine | 75 | Brunei | 15 | Guinea | 5 |
| **Brazil** | 5867 | United Arab Emirates | 365 | Algeria | 70 | Greenland | 15 | Monaco | 5 |
| Switzerland | 5527 | Hungary | 350 | Zambia | 59 | Libya | 15 | Seychelles | 5 |
| Sweden | 4826 | Slovenia | 350 | Cameroon | 58 | Puerto Rico | 15 | Andorra | 4 |
| **Ireland** | 4510 | Uruguay | 307 | Hong Kong SAR China | 58 | Benin | 13 | Bhutan | 4 |
| Finland | 4125 | Ghana | 294 | Bahrain | 55 | Barbados | 12 | French Polynesia | 4 |
| Belgium | 4070 | Thailand | 286 | Kazakhstan | 55 | Grenada | 12 | Gabon | 4 |
| Denmark | 3937 | Nepal | 282 | Cuba | 54 | Guadeloupe | 12 | Samoa | 4 |
| China | 3927 | Qatar | 275 | Bolivi | | Madagascar | 12 | Belize | 3 |
| Japan | 3597 | Romania | 262 | Cambodia | 51 | Montenegro | 12 | Bermuda | 3 |
| Norway | 3293 | Uganda | 248 | Palestinian Territories | 48 | Mozambique | 12 | Curaçao | 3 |
| Turkey | 2662 | Cyprus | 226 | Guatemala | 45 | New Caledonia | 12 | Falkland Islands | 3 |
| **Mexico** | 2361 | Luxembourg | 222 | Paraguay | 45 | Afghanistan | 11 | Palau | 3 |
| Portugal | 2190 | Croatia | 216 | Bosnia & Herzegovina | 37 | Azerbaijan | 11 | Cayman Islands | 2 |
| Austria | 2115 | Ethiopia | 213 | Senegal | 35 | El Salvador | 11 | Chad | 2 |
| South Africa | 2102 | Estonia | 209 | Moldova | 32 | Haiti | 11 | Guinea-Bissau | 2 |
| New Zealand | 2037 | Georgia | 207 | Sudan | 32 | Mauritius | 11 | Guyana | 2 |
| Israel | 2029 | Serbia | 183 | Armenia | 31 | Sierra Leone | 11 | Lesotho | 2 |
| Chile | 1631 | Iceland | 171 | Belarus | 31 | South Sudan | 11 | North Korea | 2 |
| **Argentina** | 1630 | Venezuela | 163 | Jamaica | 31 | Togo | 11 | Turkmenistan | 2 |
| Saudi Arabia | 1509 | Tanzania | 155 | Syria | 31 | Laos | 10 | Aruba | 1 |
| **Colombia** | 1476 | Morocco | 144 | Botswana | 29 | Niger | 10 | Cape Verde | 1 |
| Poland | 1339 | Sri Lanka | 143 | Fiji | 29 | Svalbard & Jan Mayen | 10 | Isle of Man | 1 |
| Singapore | 1113 | Slovakia | 141 | Gambia | 27 | Bahamas | 9 | Liberia | 1 |
| Greece | 1034 | Jordan | 118 | Côte d'Ivoire | 25 | Faroe Islands | 9 | Micronesia (Federated States of) | 1 |
| South Korea | 1004 | Vietnam | 114 | Trinidad & Tobago | 25 | Myanmar (Burma) | 9 | Montserrat | 1 |
| Pakistan | 990 | Panama | 105 | Antigua & Barbuda | 23 | French Guiana | 8 | San Marino | 1 |
| Czechia | 902 | Latvia | 102 | Congo - Kinshasa | 23 | Jersey | 7 | St. Lucia | 1 |
| Russia | 738 | Malta | 102 | Dominican Republic | 22 | Maldives | 7 | Suriname | 1 |
| Iran | 703 | Zimbabwe | 101 | Honduras | 20 | | | | |

**Figure B2: Percentage of overall Twitter followees matching users in the academic researcher list across groups.**




**Note.** Each dot represents an individual user in each group. The black rombi represent the pooled group averages. The users for the *Political Journalist* group were extracted from (Wojcieszak et al., 2022). The *Science Journalist* users correspond to a curated science journalist list on Twitter (ID=1186721173222100994).

**Figure B3: Google Topic interest of early COVID related terms**

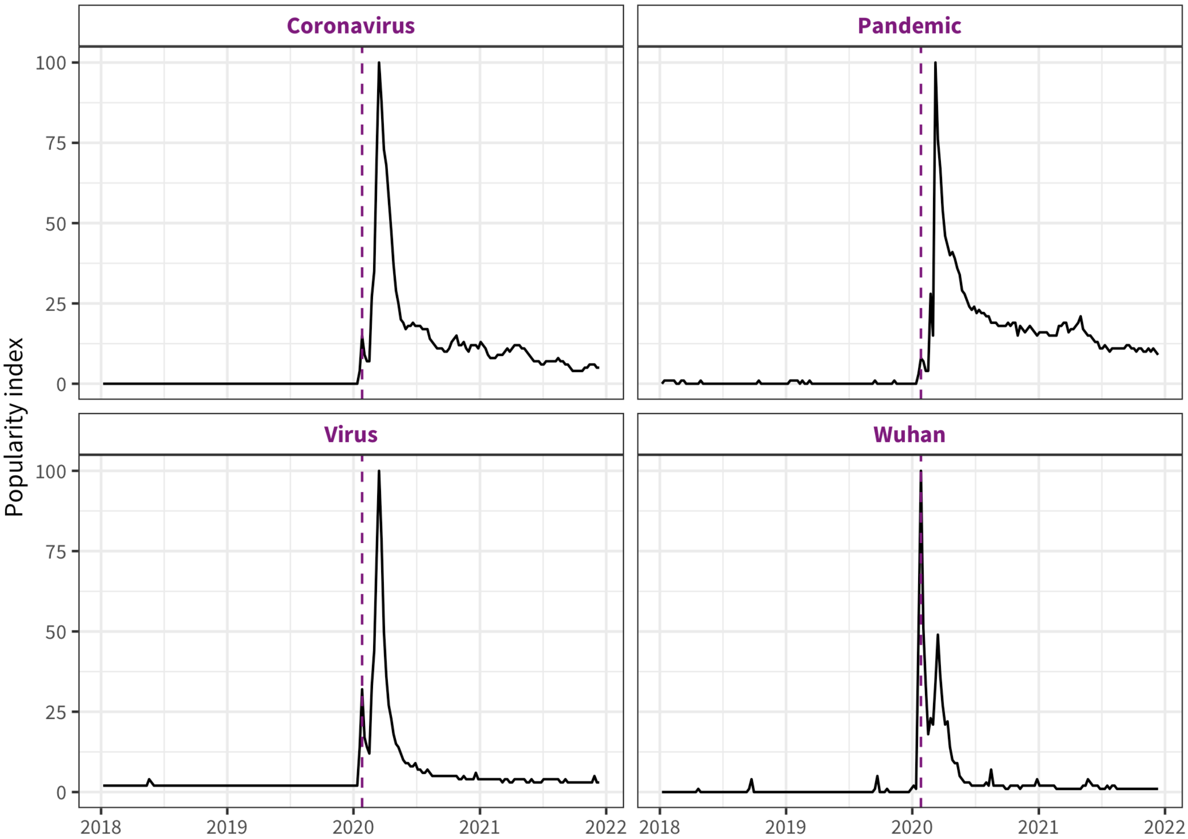


**Table B4:** Academic researchers with the largest increase in legislator followers following the declaration of COVID

| Twitter handle | Name | Scientific branch | Last known affiliation | Base | Gained | $\Delta$ increase |
|---|---|---|---|---|---|---|
| @c_drosten | Christian Drosten | Natural science | Humboldt-Universität zu Berlin | 52 | 102 | 1.96x |
| @DrTedros | Tedros Adhanom Ghebreyesus | Applied science | World Health Organization | 24 | 52 | 2.17x |
| @uksciencechief | Patrick Vallance | Applied science | GlaxoSmithKline | 35 | 50 | 1.43x |
| @devisridhar | Devi Sridhar | Applied science | University of Edinburgh | 5 | 27 | 5.4x |
| @hendrikstreeck | Hendrik Streeck | No category | — | 16 | 23 | 1.44x |
| @d_spiegel | David Spiegelhalter | Formal science | University of Cambridge | 1 | 18 | 18x |
| @ronan_glynn | Robert J. Glynn | Applied science | Brigham and Women's Hospital | 8 | 16 | 2x |
| @jasonleitch | J. Leitch | Social science | Scottish Government | 5 | 15 | 3x |
| @GabrielScally | Gabriel Scally | Applied science | University of Bristol | 1 | 15 | 15x |
| @globalhlthtwit | Anthony Costello | Applied science | UCL Institute of Child Health | 3 | 15 | 5x |
| @JeremyFarrar | Jeremy Farrar | Applied science | Wellcome Trust | 4 | 14 | 3.5x |
| @oriolmitja | Oriol Mitjà | Applied science | Fight AIDS Foundation | 0 | 14 | — |
| @anandMenon1 | Anand Menon | Social science | Innovate UK | 8 | 13 | 1.62x |
| @DrEricDing | Eric L. Ding | Applied science | Microclinic International | 3 | 13 | 4.33x |
| @adamjkucharski | Adam J. Kucharski | Applied science | London School of Hygiene & Tropical Medicine | 10 | 13 | 1.3x |
| @claire_ainsley | Claire Ainsley | Formal science | — | 0 | 12 | — |
| @mlipsitch | Marc Lipsitch | Applied science | Harvard University | 4 | 12 | 3x |
| @CiesekSandra | Sandra Ciesek | Applied science | German Center for Infection Research | 0 | 11 | — |
| @CathCalderwood1 | Catherine Calderwood | Applied science | Scottish Government | 1 | 11 | 11x |
| @ASlavitt | Andrew Slavitt | Applied science | Centers for Medicare and Medicaid Services | 4 | 10 | 2.5x |
| @pia_lamberty | Pia Lamberty | Social science | Johannes Gutenberg University of Mainz | 0 | 10 | — |
| @adam_tooze | Adam Tooze | Social science | Columbia University | 4 | 10 | 2.5x |
| @GrimmVeronika | Veronika Grimm | Social science | University of Erlangen-Nuremberg | 4 | 10 | 2.5x |
| @hans_kluge | Hans Kluge | Applied science | World Health Organization Regional Office for Europe | 0 | 10 | — |
| @miotei | Miguel Otero-Iglesias | Social science | Real Instituto Elcano | 0 | 9 | — |

**Table B5:** Academic researchers with the most retweets by legislators following the declaration of COVID

| Twitter handle | Name | Scientific branch | Last known affiliation | # pre | # post | # legs |
|---|---|---|---|---|---|---|
| @G_Caballero_M | Gonzalo Caballero-Miguez | Social science | Universidade de Vigo | 0 | 261 | 11 |
| @TorstenBell | Torsten Bell | Social science | — | 0 | 189 | 53 |
| @DrTedros | Tedros Adhanom Ghebreyesus | Applied science | World Health Organization | 0 | 109 | 78 |
| @jasonleitch | J. Leitch | Social science | Scottish Government | 0 | 108 | 28 |
| @josepcosta | Josep Costa | Social science | — | 0 | 105 | 8 |
| @perezlozano | Lluis Perez-Lozano | Social science | Pompeu Fabra University | 0 | 88 | 10 |
| @cblackst | Cindy Blackstock | Social science | McGill University | 0 | 81 | 12 |
| @premnsikka | Prem Sikka | Applied science | University of Essex | 0 | 74 | 9 |
| @janephilpott | Jane Philpott | Applied science | University of Toronto | 0 | 74 | 18 |
| @Orla_Hegarty | Orla Hegarty | Social science | University College Dublin | 0 | 69 | 12 |
| @martamartirio | Marta Martín-Llaguno | Social science | University of Alicante | 0 | 69 | 11 |
| @c_drosten | Christian Drosten | Natural science | Humboldt-Universität zu Berlin | 0 | 68 | 43 |
| @XSalaimartin | Xavier Sala-i-Martin | Social science | MBIA | 0 | 66 | 6 |
| @juanrallo | Juan Ramón Rallo | Humanities | IE University | 0 | 63 | 28 |
| @RAWnGreen | Julia K. Green | Natural science | University of California, Berkeley | 17 | 63 | 1 |
| @uksciencechief | Patrick Vallance | Applied science | GlaxoSmithKline | 0 | 62 | 40 |
| @doctor_oxford | Rachel Clarke | Applied science | University of Oxford | 0 | 56 | 30 |
| @RZitelmann | Rainer Zitelmann | Social science | National Coalition of Independent Scholars | 0 | 55 | 6 |
| @ronan_glynn | Robert J. Glynn | Applied science | Brigham and Women's Hospital | 0 | 55 | 18 |
| @ZulmaCucunuba | Zulma M. Cucunubá | Applied science | Imperial College London | 0 | 52 | 4 |
| @JaimePalomera | Jaime Palomera | Social science | University of Barcelona | 0 | 51 | 13 |
| @GabrielScally | Gabriel Scally | Applied science | University of Bristol | 0 | 49 | 22 |
| @schnellenbachj | Jan Schnellenbach | Social science | Brandenburg University of Technology | 0 | 49 | 3 |
| @gebelque | Germà Bel | Social science | University of Barcelona | 0 | 44 | 5 |
| @jaumepadros | J. Padrós-Selma | Social science | West Virginia University College of Law | 0 | 44 | 11 |

**Table B6:** Academic researchers with the most mentions by legislators following the declaration of COVID

| Twitter handle | Name | Scientific branch | Last known affiliation | # pre | # post | # legs |
|---|---|---|---|---|---|---|
| @G_Caballero_M | Gonzalo Caballero-Miguez | Social science | Universidade de Vigo | 0 | 52 | 9 |
| @DrTedros | Tedros Adhanom Ghebreyesus | Applied science | World Health Organization | 0 | 25 | 13 |
| @huw4ogmore | Harriet Harden-Davies | Social science | University of Wollongong | 1 | 20 | 1 |
| @CDCDirector | Rochelle P Walensky | Social science | Centers for Disease Control and Prevention | 0 | 19 | 15 |
| @JoMalagon | Jonathan Malagon | No category | — | 0 | 19 | 11 |
| @c_drosten | Christian Drosten | Natural science | Humboldt-Universität zu Berlin | 0 | 14 | 14 |
| @SteveFDA | Stephen M. Hahn | Applied science | Annenberg Public Policy Center | 0 | 14 | 10 |
| @Martin_M_Guzman | Martin Guzman | Social science | Columbia University | 0 | 12 | 8 |
| @ilariacapua | Ilaria Capua | Applied science | University of Florida Health | 0 | 11 | 5 |
| @samuel_garcias | — | No category | — | 0 | 10 | 8 |
| @ASlavitt | Andrew Slavitt | Applied science | Centers for Medicare and Medicaid Services | 0 | 8 | 3 |
| @jasonleitch | J. Leitch | Social science | Scottish Government | 0 | 8 | 6 |
| @uksciencechief | Patrick Vallance | Applied science | GlaxoSmithKline | 0 | 8 | 8 |
| @ashishkjha | Ashish K. Jha | Applied science | Brown University | 0 | 7 | 1 |
| @JanezPotocnik22 | Janez Potočnik | Social science | United Nations Environment Programme | 0 | 6 | 1 |
| @DrJV75 | Justin Varney | Applied science | Public Health England | 3 | 6 | 1 |
| @hendrikstreeck | — | No category | — | 0 | 6 | 6 |
| @gabriel_zucman | Gabriel Zucman | Social science | University of California, Berkeley | 0 | 6 | 3 |
| @WRicciardi | — | No category | — | 0 | 6 | 4 |
| @Healthmac | Christopher Mackie | Applied science | Middlesex London Health Unit | 0 | 6 | 3 |
| @AlanDersh | Alan M. Dershowitz | Social science | — | 0 | 6 | 2 |
| @CarloMasala1 | Carlo Masala | Social science | Bundeswehr University Munich | 0 | 6 | 3 |
| @GaviSeth | Seth Berkley | Applied science | Gavi | 0 | 6 | 4 |
| @lugaricano | Luis Garicano | Social science | IE University | 0 | 5 | 4 |
| @MarvinJRees | Marvin Rees | Social science | — | 0 | 5 | 4 |

**Table B7:** Academic researchers with the most quotes by legislators following the declaration of COVID

| Twitter handle | Name | Scientific branch | Last known affiliation | # pre | # post | # legs |
|---|---|---|---|---|---|---|
| @TorstenBell | Torsten Bell | Social science | — | 0 | 33 | 31 |
| @janephilpott | Jane Philpott | Applied science | University of Toronto | 0 | 23 | 17 |
| @c_drosten | Christian Drosten | Natural science | Humboldt-Universität zu Berlin | 0 | 22 | 10 |
| @DrTedros | Tedros Adhanom Ghebreyesus | Applied science | World Health Organization | 0 | 20 | 18 |
| @G_Caballero_M | Gonzalo Caballero-Miguez | Social science | Universidade de Vigo | 8 | 17 | 2 |
| @hans_kluge | Hans Kluge | Applied science | World Health Organization Regional Office for Europe | 0 | 14 | 13 |
| @ASlavitt | Andrew Slavitt | Applied science | Centers for Medicare and Medicaid Services | 0 | 13 | 10 |
| @juanrallo | Juan Ramón Rallo | Humanities | IE University | 0 | 11 | 8 |
| @Orla_Hegarty | Orla Hegarty | Social science | University College Dublin | 0 | 10 | 4 |
| @cblackst | Cindy Blackstock | Social science | McGill University | 9 | 10 | 3 |
| @jdportes | Jonathan Portes | Social science | King's College London | 0 | 9 | 5 |
| @doctor_oxford | Rachel Clarke | Applied science | University of Oxford | 7 | 9 | 6 |
| @AlanDersh | Alan M. Dershowitz | Social science | — | 0 | 8 | 4 |
| @ronan_glynn | Robert J. Glynn | Applied science | Brigham and Women's Hospital | 0 | 8 | 5 |
| @ianbremmer | Ian A. (Ian Arthur) Bremmer | Social science | Hoover Institution | 0 | 8 | 8 |
| @MacaesBruno | Bruno Macaes | Social science | — | 0 | 7 | 2 |
| @GabrielScally | Gabriel Scally | Applied science | University of Bristol | 0 | 7 | 4 |
| @uksciencechief | Patrick Vallance | Applied science | GlaxoSmithKline | 0 | 7 | 6 |
| @nntaleb | Nassim Nicholas Taleb | Formal science | New York University | 0 | 7 | 3 |
| @oriolmitja | Oriol Mitjà | Applied science | Fight AIDS Foundation | 0 | 7 | 3 |
| @Miguel__Lorente | Miguel Lorente-Acosta | Humanities | University of Granada | 5 | 7 | 1 |
| @MaxCRoser | Max Roser | Social science | Center for Global Development | 0 | 7 | 5 |
| @jasonleitch | J. Leitch | Social science | Scottish Government | 0 | 6 | 6 |
| @premnsikka | Prem Sikka | Applied science | University of Essex | 0 | 6 | 6 |
| @jsuedekum | Jens Suedekum | Social science | Heinrich Heine University Düsseldorf | 0 | 6 | 5 |

**Table B8:** Academic researchers with the most replies by legislators following the declaration of COVID

| Twitter handle | Name | Scientific branch | Last known affiliation | # pre | # post | # legs |
|---|---|---|---|---|---|---|
| @CarloMasala1 | Carlo Masala | Social science | Bundeswehr University Munich | 36 | 69 | 10 |
| @BachmannRudi | Ruediger Bachmann | Social science | University of Notre Dame | 0 | 33 | 4 |
| @SDullien | Sebastian Dullien | Social science | Hans Böckler Foundation | 0 | 29 | 6 |
| @mquijoux | Maxime Quijoux | Social science | Laboratoire Interdisciplinaire pour la Sociologie Economique | 6 | 21 | 1 |
| @StephanieCarvin | Stephanie Carvin | Social science | University of Ottawa | 0 | 21 | 5 |
| @thesismum | Anja Katharina Peters | Social science | — | 0 | 17 | 4 |
| @AnMailleach | Eoin O'Malley | Social science | Dublin City University | 14 | 16 | 9 |
| @AchimTruger | Achim Truger | Social science | University of Duisburg-Essen | 0 | 16 | 5 |
| @jsuedekum | Jens Suedekum | Social science | Heinrich Heine University Düsseldorf | 0 | 15 | 3 |
| @m_kubiciel | Michael Kubiciel | Social science | University of Cologne | 0 | 15 | 4 |
| @DrGrandMal | Dennis Müller | Natural science | Heidelberg University | 0 | 14 | 5 |
| @HerrLuehmann | Michael Lühmann | Natural science | — | 0 | 13 | 4 |
| @Lars_Feld | Lars P. Feld | Social science | Walter Eucken Institut | 0 | 13 | 7 |
| @gavinjdaly | Gavin Daly | Social science | National University of Ireland, Maynooth | 0 | 11 | 4 |
| @tatterededge | Lelainia Lloyd | Social science | — | 4 | 11 | 1 |
| @PolProfSteve | Steven Fielding | Social science | University of Nottingham | 0 | 11 | 7 |
| @lofferg | Christopher Gohl | Social science | Leibniz-Institut für Wissensmedien | 0 | 10 | 3 |
| @JuergenZimmerer | Jürgen Zimmerer | Social science | University of Sheffield | 2 | 10 | 1 |
| @wargonm | Mathias Wargon | Applied science | Centre Hospitalier Saint-Denis | 0 | 10 | 5 |
| @devisridhar | Devi Sridhar | Applied science | University of Edinburgh | 0 | 9 | 2 |
| @DannyFiler | Danny Filer | Applied science | University College London | 0 | 9 | 3 |
| @cataperezcorrea | Catalina Pérez Correa | Social science | University of Iceland | 1 | 9 | 1 |
| @Puettmann_Bonn | Andreas Püttmann | Social science | — | 0 | 9 | 4 |
| @drcrouchback | Paul W Keeley | Applied science | University of Glasgow | 1 | 9 | 1 |
| @T_Ortelt | Tobias R. Ortelt | Formal science | TU Dortmund University | 0 | 9 | 4 |

**Table B9:** Political parties and families by country

| Country | Party | Family | Country | Party | Family |
|---|---|---|---|---|---|
| Argentina | Frente PRO | CON | France | DVG | SOC |
| Argentina | Unión Cívica Radical | SOC | France | POI | LEF |
| Brazil | MDB | SOC | France | Horizons-CCB | CON |
| Brazil | UNIÃO | DIV | France | LR-Nouvelle énergie | CON |
| Brazil | PL | NAT | France | DVG | ETH |
| Brazil | PSD | LIB | France | UDI | CON |
| Brazil | REDE (PSOL REDE) | ECO | France | Tavini | LEF |
| Brazil | PSDB (PSDB Cidadania) | CHR | France | LREM-PE | LIB |
| Brazil | PT (FE Brasil) | LEF | France | LREM-GNC | LIB |
| Brazil | PODE | SIP | France | LREM, Cap21 | LIB |
| Brazil | Republicanos | CON | France | GUSR | ETH |
| Brazil | PDT | SOC | France | PRG | MI |
| Brazil | Cidadania (PSDB Cidadania) | CHR | France | LREM - TdP | LIB |
| Brazil | PSB | SOC | France | PNC | ETH |
| Brazil | PP | CON | France | REG | ETH |
| Brazil | PSC | NAT | France | PS | SOC |
| Brazil | PROS | DIV | France | DVC | LIB |
| Canada | Liberal | LIB | France | LREM, EC | LIB |
| Canada | New Democratic | SOC | France | DVD | MI |
| Canada | Green | ECO | Germany | CDU/CSU (CDU) | CHR |
| Canada | Conservative | CON | Germany | Grüne | ECO |
| Canada | Bloc Québécois | SIP | Germany | FDP | LIB |
| Canada | Independent | MI | Germany | SPD | SOC |
| Colombia | CD | CON | Germany | Linke | LEF |
| Colombia | MAIS | ETH | Germany | CDU/CSU (CSU) | CHR |
| Colombia | AV | ECO | Germany | AfD | NAT |
| Colombia | PUG | SIP | Germany | Fraktionslos | MI |
| Colombia | CH | LEF | Ireland | Fianna Fáil | CON |
| Colombia | CR | LIB | Ireland | Fine Gael | CHR |
| Colombia | MAIS | ETH | Ireland | Sinn Féin | SIP |
| Colombia | UP | LEF | Ireland | Green Party | ECO |
| Colombia | PLC | LIB | Ireland | Independent | MI |
| Colombia | PDA | LEF | Ireland | Solidarity–People Before Profit | LEF |
| Colombia | MIRA | NAT | Ireland | Labour | SOC |
| Colombia | AV | ECO | Ireland | Aontú | CHR |
| Colombia | ADA | ETH | Ireland | Social Democrats | SOC |
| Colombia | ASI | ETH | Ireland | Independents 4 Change | DIV |
| Colombia | COM | LEF | Italy | PARTITO DEMOCRATICO - ITALIA DEMOCRATICA E PROGRESSISTA | SOC |
| Colombia | VO | ECO | Italy | FRATELLI D'ITALIA | CON |
| Colombia | PC | CON | Italy | LEGA - SALVINI PREMIER | NAT |
| Colombia | ASI | ETH | Italy | AZIONE - ITALIA VIVA - RENEW EUROPE | DIV |
| Colombia | CJL | NAT | Italy | FORZA ITALIA - BERLUSCONI PRESIDENTE - PPE | CON |
| Colombia | AICO | ETH | Italy | MISTO-+EUROPA | SIP |
| France | LREM | LIB | Italy | MOVIMENTO 5 STELLE | SIP |
| France | G.s | ECO | Italy | NOI MODERATI (NOI CON L'ITALIA, CORAGGIO ITALIA, UDC, ITALIA AL CENTRO)-MAIE | DIV |
| France | TdP | LIB | Italy | ALLEANZA VERDI E SINISTRA | ECO |
| France | DVD | LIB | Italy | MISTO-MINORANZE LINGUISTICHE | ETH |
| France | LR | CON | Mexico | PAN | CON |
| France | RN | NAT | Mexico | Morena | LEF |
| France | MoDem | LIB | Mexico | Sin Partido | MI |
| France | LFI | LEF | Mexico | PRI | SOC |
| France | DVC | LIB | Mexico | Partido Movimiento Ciudadano | SOC |
| France | EELV | ECO | Mexico | Partido Encuentro Social | CON |
| France | Horizons | CON | Mexico | PRD | SOC |
| France | PS | SOC | Mexico | Partido Verde | ECO |
| France | PCF | LEF | Mexico | Worker's Party | LEF |
| France | LREM-TdP | LIB | Mexico | Movimiento Ciudadano | SOC |
| France | LND | ECO | Spain | PSOE | SOC |
| France | PRV | LIB | Spain | JxCat-JUNTS (Junts) | ETH |
| France | LREM-EC | LIB | Spain | EC-UP | ETH |
| France | PRV | ETH | Spain | PSC-PSOE | SOC |
| France | Agir | LIB | Spain | Vox | NAT |
| France | DVC | ETH | Spain | ERC-S | ETH |
| France | FP | LIB | Spain | PSE-EE-PSOE | SOC |
| France | DVG | MI | Spain | PsdeG-PSOE | SOC |
| France | LC | ETH | Spain | EH Bildu | ETH |
| France | LFI-E! | LEF | Spain | UP | ETH |
| France | REV | LEF | Spain | PP | CON |
| France | FaC | ETH | Spain | ECP-GUAYEM EL CANVI | ETH |
| France | MDES | LEF | Spain | MÁS PAÍS-EQUO | LEF |
| France | EÉLV | ECO | Spain | NA+ | ETH |
| France | PPDG | SOC | Spain | EAJ-PNV | ETH |
| France | LREM | LIB | Spain | JxCat-JUNTS (PDeCAT) | ETH |

**Table B9:** Political parties and families by country

| Country | Party | Family | Country | Party | Family |
|---|---|---|---|---|---|
| France | Agir | CON | Spain | Cs | LIB |
| France | LR | CON | Spain | CUP-PR | ETH |
| France | DVG | LEF | Spain | MÉS COMPROMÍS | ETH |
| France | GE | ECO | Spain | BNG | ETH |
| France | DVD | CON | Spain | CCa-PNC-NC | ETH |
| France | CE | LIB | Spain | UP | ETH |
| France | DLF | MI | Spain | Cs | LIB |
| France | LR-A droite ! | CON | Spain | PP-FORO | CON |
| France | RR | LIB | Spain | NC-CCa-PNC | ETH |
| France | FGPS | SOC | Spain | PP-FORO | CON |
| France | EXD | MI | Spain | PRC | ETH |
| France | LFI-PG | LEF | UK | Labour Co-operative | SOC |
| France | LR-SL | CON | UK | Conservative | CON |
| France | DVD | LIB | UK | Labour | SOC |
| France | DVD | ETH | UK | Liberal Democrats | LIB |
| France | PRV | LIB | UK | Sinn Féin | LEF |
| France | LS | NAT | UK | Scottish National | ETH |
| France | LFI-PD | LEF | UK | Green | ECO |
| France | LREM-RSM | LIB | UK | SDLP | SOC |
| France | PLR | LEF | UK | Plaid Cymru | ETH |
| France | UDI | ETH | UK | DUP | ETH |
| France | Horizons | LIB | UK | Alliance | LIB |
| France | Horizons-LREM | CON | UK | Speaker | CON |
| France | LREM | CON | UK | SNP | ETH |
| France | LFI-RÉ974 | LEF | US | Democratic | SOC |
| France | PRV-LREM | LIB | US | Republican | CON |
| France | PPM | SOC | US | Independent | MI |

**Table B10:** Estimated effects of legislator and legislature characteristics on the proportion of researchers across behaviors. (*Fig. 3* in main body and *Figs. C1-C5* )

| | Following | Mentioning | Retweeting | Quote tweeting | Replying | Liking |
|---|---|---|---|---|---|---|
| Doctoral studies | 0.009*** | 0.007*** | 0.006*** | 0.006*** | 0.008*** | 0.007*** |
| | [0.008, 0.011] | [0.005, 0.008] | [0.005, 0.008] | [0.004, 0.008] | [0.005, 0.011] | [0.005, 0.010] |
| Female | 0.000 | 0.000 | −0.001 | −0.001 | 0.001 | 0.000 |
| | [−0.001, 0.001] | [−0.001, 0.001] | [−0.002, 0.000] | [−0.002, 0.001] | [−0.001, 0.003] | [−0.002, 0.001] |
| Age: 35-44 (ref: Under 35) | 0.000 | 0.001 | −0.001 | 0.001 | 0.001 | 0.001 |
| | [−0.002, 0.001] | [−0.001, 0.003] | [−0.003, 0.001] | [−0.001, 0.004] | [−0.002, 0.005] | [−0.002, 0.004] |
| 45-54 | −0.003** | 0.000 | −0.003** | 0.000 | 0.001 | −0.001 |
| | [−0.005, −0.001] | [−0.002, 0.001] | [−0.005, −0.001] | [−0.002, 0.003] | [−0.003, 0.004] | [−0.004, 0.002] |
| 55-64 | −0.003** | 0.000 | −0.002+ | 0.002 | 0.000 | 0.001 |
| | [−0.004, −0.001] | [−0.002, 0.002] | [−0.004, 0.000] | [−0.001, 0.004] | [−0.003, 0.003] | [−0.002, 0.003] |
| 65-74 | −0.004*** | 0.000 | −0.003** | 0.001 | −0.001 | −0.001 |
| | [−0.006, −0.002] | [−0.002, 0.002] | [−0.005, −0.001] | [−0.002, 0.003] | [−0.005, 0.003] | [−0.004, 0.002] |
| above 75 | −0.005** | −0.001 | −0.004* | 0.000 | −0.002 | −0.003 |
| | [−0.008, −0.002] | [−0.004, 0.002] | [−0.008, −0.001] | [−0.004, 0.004] | [−0.007, 0.003] | [−0.008, 0.002] |
| Party family: Conservative (ref: Other) | −0.003*** | −0.001 | −0.003** | −0.003* | −0.003+ | −0.003* |
| | [−0.005, −0.002] | [−0.002, 0.001] | [−0.004, −0.001] | [−0.005, 0.000] | [−0.006, 0.000] | [−0.006, −0.001] |
| Ecological | 0.008*** | 0.002 | 0.004** | 0.002 | 0.003 | 0.004* |
| | [0.005, 0.010] | [−0.001, 0.004] | [0.001, 0.007] | [−0.002, 0.005] | [−0.001, 0.008] | [0.000, 0.008] |
| Liberal | 0.001 | 0.000 | −0.001 | −0.002 | 0.004* | 0.002 |
| | [−0.001, 0.003] | [−0.002, 0.002] | [−0.003, 0.001] | [−0.004, 0.001] | [0.000, 0.007] | [−0.001, 0.005] |
| Nationalistic and radical right | −0.005*** | −0.002 | −0.001 | −0.003+ | −0.006** | −0.005** |
| | [−0.007, −0.003] | [−0.004, 0.001] | [−0.004, 0.001] | [−0.006, 0.000] | [−0.009, −0.002] | [−0.008, −0.001] |
| Social democratic | 0.003** | 0.002+ | 0.001 | 0.001 | 0.000 | 0.002 |
| | [0.001, 0.004] | [0.000, 0.003] | [−0.001, 0.003] | [−0.002, 0.003] | [−0.003, 0.003] | [−0.001, 0.005] |
| Socialist and other left | 0.000 | −0.001 | 0.000 | −0.003+ | −0.001 | 0.000 |
| | [−0.002, 0.003] | [−0.003, 0.001] | [−0.003, 0.002] | [−0.006, 0.000] | [−0.005, 0.003] | [−0.004, 0.004] |
| Followers qtl: Q2 (ref: Q1) | 0.000 | 0.001 | 0.000 | 0.001 | −0.003* | −0.001 |
| | [−0.001, 0.002] | [−0.001, 0.002] | [−0.001, 0.002] | [0.000, 0.003] | [−0.005, 0.000] | [−0.003, 0.001] |
| Q3 | 0.004*** | 0.003*** | 0.003*** | 0.003** | 0.000 | 0.004*** |
| | [0.002, 0.005] | [0.002, 0.005] | [0.002, 0.005] | [0.001, 0.005] | [−0.003, 0.002] | [0.002, 0.007] |
| Q4 | 0.007*** | 0.005*** | 0.003*** | 0.005*** | 0.000 | 0.006*** |
| | [0.005, 0.008] | [0.003, 0.006] | [0.002, 0.005] | [0.003, 0.007] | [−0.003, 0.002] | [0.003, 0.008] |
| Friends qtl: Q2 (ref: Q1) | 0.001* | 0.000 | 0.000 | −0.002* | 0.001 | 0.000 |
| | [0.000, 0.003] | [−0.001, 0.002] | [−0.002, 0.001] | [−0.004, 0.000] | [−0.001, 0.003] | [−0.002, 0.002] |
| Q3 | 0.002*** | 0.001 | 0.000 | −0.001 | 0.002* | 0.001 |
| | [0.001, 0.004] | [0.000, 0.002] | [−0.001, 0.002] | [−0.003, 0.001] | [0.000, 0.005] | [−0.001, 0.004] |
| Q4 | 0.003*** | 0.001 | 0.002* | 0.000 | 0.003* | 0.003** |
| | [0.001, 0.004] | [−0.001, 0.002] | [0.000, 0.003] | [−0.002, 0.002] | [0.000, 0.005] | [0.001, 0.006] |
| Region: LAC (ref: EUR) | −0.002 | 0.001 | −0.002 | −0.001 | −0.003 | −0.003 |
| | [−0.009, 0.005] | [−0.004, 0.006] | [−0.009, 0.004] | [−0.007, 0.006] | [−0.010, 0.003] | [−0.012, 0.006] |
| NAm | −0.002 | −0.001 | 0.000 | 0.000 | −0.002 | 0.000 |
| | [−0.009, 0.004] | [−0.005, 0.004] | [−0.005, 0.006] | [−0.006, 0.005] | [−0.008, 0.003] | [−0.008, 0.008] |
| Parliamentary republic | 0.010** | 0.005* | 0.007* | 0.007* | 0.008** | 0.010* |
| | [0.003, 0.016] | [0.000, 0.009] | [0.001, 0.012] | [0.001, 0.012] | [0.003, 0.013] | [0.002, 0.018] |
| SD (Observations) | 0.013 | 0.013 | 0.014 | 0.017 | 0.022 | 0.019 |
| Num.Obs. | 3381 | 3318 | 3335 | 3145 | 3311 | 3295 |
| R2 Marg. | 0.252 | 0.082 | 0.109 | 0.068 | 0.075 | 0.117 |
| R2 Cond. | 0.308 | 0.119 | 0.155 | 0.101 | 0.092 | 0.161 |
| ICC | 0.1 | 0.0 | 0.1 | 0.0 | 0.0 | 0.0 |
| RMSE | 0.01 | 0.01 | 0.01 | 0.02 | 0.02 | 0.02 |

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

**Table B11: Pooled scholars.** Results from a logistic mixed effects model with country of legislature random effects on following and engagement with academic researchers during the COVID versus pre-COVID periods with a 12 week bandwidth (*Fig. 5a* in main body)

| | Following | Mentioning | Retweeting | Quote tweeting | Replying |
|---|---|---|---|---|---|
| Post-COVID | $0.343^{***}$ | $0.306^{***}$ | $0.162^{***}$ | $0.185^{***}$ | $-0.052$ |
| | [0.296, 0.390] | [0.202, 0.411] | [0.126, 0.198] | [0.096, 0.274] | [−0.115, 0.011] |
| Num.Obs. | 334 670 | 315 542 | 925 499 | 166 371 | 266 846 |
| R2 Marg. | 0.008 | 0.007 | 0.002 | 0.002 | 0.000 |
| R2 Cond. | 0.097 | 0.054 | 0.145 | 0.075 | 0.256 |
| ICC | 0.1 | 0.0 | 0.1 | 0.1 | 0.3 |
| RMSE | 0.15 | 0.07 | 0.11 | 0.11 | 0.12 |

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

**Table B12: Following (by branch).** Results from a logistic mixed effects model with country of legislature random effects on following and engagement with academic researchers during the COVID versus pre-COVID periods with a 12 week bandwidth (*Fig. 5b* in main body)

| | Medical science | Humanities | Formal science | Natural science | Social science | No category |
|---|---|---|---|---|---|---|
| Post-COVID | $0.716^{***}$ | $0.368^{***}$ | $0.447^{***}$ | $0.286^{***}$ | $0.186^{***}$ | $0.320^{**}$ |
| | [0.608, 0.824] | [0.161, 0.575] | [0.253, 0.640] | [0.153, 0.418] | [0.121, 0.251] | [0.114, 0.526] |
| Num.Obs. | 334 670 | 334 670 | 334 670 | 334 670 | 334 670 | 334 670 |
| R2 Marg. | 0.031 | 0.009 | 0.014 | 0.006 | 0.002 | 0.007 |
| R2 Cond. | 0.195 | 0.119 | 0.081 | 0.089 | 0.101 | 0.125 |
| ICC | 0.2 | 0.1 | 0.1 | 0.1 | 0.1 | 0.1 |
| RMSE | 0.07 | 0.03 | 0.04 | 0.05 | 0.10 | 0.03 |

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

**Table B13: Mentioning (by branch).** Results from a logistic mixed effects model with country of legislature random effects on following and engagement with academic researchers during the COVID versus pre-COVID periods with a 12 week bandwidth (*Fig. 5b* in main body)

| | Medical science | Humanities | Formal science | Natural science | Social science | No category |
|---|---|---|---|---|---|---|
| Post-COVID | $1.124^{***}$ | $0.135$ | $-0.327$ | $0.355+$ | $0.263^{***}$ | $-0.334+$ |
| | [0.825, 1.422] | [−0.361, 0.630] | [−0.784, 0.129] | [−0.022, 0.732] | [0.131, 0.396] | [−0.680, 0.012] |
| Num.Obs. | 315 542 | 315 542 | 315 542 | 315 542 | 315 542 | 315 542 |
| R2 Marg. | 0.077 | 0.001 | 0.006 | 0.008 | 0.005 | 0.006 |
| R2 Cond. | 0.194 | 0.366 | 0.273 | 0.208 | 0.104 | 0.271 |
| ICC | 0.1 | 0.4 | 0.3 | 0.2 | 0.1 | 0.3 |
| RMSE | 0.03 | 0.01 | 0.01 | 0.02 | 0.05 | 0.02 |

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

**Table B14: Retweeting (by branch).** Results from a logistic mixed effects model with country of legislature random effects on following and engagement with academic researchers during the COVID versus pre-COVID periods with a 12 week bandwidth (*Fig. 5b* in main body)

| | Medical science | Humanities | Formal science | Natural science | Social science | No category |
|---|---|---|---|---|---|---|
| Post-COVID | $0.842^{***}$ | $-0.068$ | $-0.288^{***}$ | $0.251^{***}$ | $0.038+$ | $0.030$ |
| | [0.741, 0.942] | [−0.237, 0.101] | [−0.452, −0.123] | [0.130, 0.372] | [−0.007, 0.084] | [−0.150, 0.211] |
| Num.Obs. | 925 499 | 925 499 | 925 499 | 925 499 | 925 499 | 925 499 |
| R2 Marg. | 0.041 | 0.000 | 0.005 | 0.004 | 0.000 | 0.000 |
| R2 Cond. | 0.219 | 0.198 | 0.253 | 0.237 | 0.141 | 0.175 |
| ICC | 0.2 | 0.2 | 0.2 | 0.2 | 0.1 | 0.2 |
| RMSE | 0.05 | 0.02 | 0.02 | 0.03 | 0.09 | 0.02 |

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

**Table B15: Quote tweeting (by branch).** Results from a logistic mixed effects model with country of legislature random effects on following and engagement with academic researchers during the COVID versus pre-COVID periods with a 12 week bandwidth (*Fig. 5b* in main body)

| | Medical science | Humanities | Formal science | Natural science | Social science | No category |
|---|---|---|---|---|---|---|
| Post-COVID | $0.790^{***}$ | $-0.021$ | $-0.061$ | $0.367^{*}$ | $0.012$ | $0.498^{*}$ |
| | [0.552, 1.027] | [−0.427, 0.386] | [−0.424, 0.302] | [0.075, 0.658] | [−0.101, 0.126] | [0.048, 0.948] |
| Num.Obs. | 166 371 | 166 371 | 166 371 | 166 371 | 166 371 | 166 371 |
| R2 Marg. | 0.039 | 0.000 | 0.000 | 0.008 | 0.000 | 0.016 |
| R2 Cond. | 0.163 | 0.268 | 0.273 | 0.183 | 0.067 | 0.137 |
| ICC | 0.1 | 0.3 | 0.3 | 0.2 | 0.1 | 0.1 |
| RMSE | 0.05 | 0.02 | 0.03 | 0.03 | 0.09 | 0.02 |

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

**Table B16: Replying tweeting (by branch).** Results from a logistic mixed effects model with country of legislature random effects on following and engagement with academic researchers during the COVID versus pre-COVID periods with a 12 week bandwidth (*Fig. 5b* in main body)

| | Medical science | Humanities | Formal science | Natural science | Social science | No category |
|---|---|---|---|---|---|---|
| Post-COVID | $0.187+$ | $-0.249^{*}$ | $-0.304^{**}$ | $-0.227^{*}$ | $-0.028$ | $0.309+$ |
| | [−0.003, 0.377] | [−0.494, −0.003] | [−0.534, −0.074] | [−0.406, −0.049] | [−0.110, 0.053] | [−0.004, 0.621] |
| Num.Obs. | 266 846 | 266 846 | 266 846 | 266 846 | 266 846 | 266 846 |
| R2 Marg. | 0.002 | 0.003 | 0.005 | 0.003 | 0.000 | 0.005 |
| R2 Cond. | 0.309 | 0.248 | 0.265 | 0.311 | 0.271 | 0.304 |
| ICC | 0.3 | 0.2 | 0.3 | 0.3 | 0.3 | 0.3 |
| RMSE | 0.04 | 0.03 | 0.03 | 0.04 | 0.09 | 0.02 |

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

# Appendix C Supporting analyses

## C.1 Estimated effects of legislator and legislature characteristics on the proportion of researchers on behaviors

These are results from linear mixed-effects models with country of legislature random effects with age (under 35), party family (other), country region (Europe), system (presidential), and Q1 for followers and friends as references for categorical variables. Panel a presents the coefficients with 80% and 95% confidence intervals. The conditional predictions are computed with numeric covariates are held at their means and the other covariates at their modes: no research degree, presidential, European, male, 45-54, Conservative party, and the first quartile of followers and friends.

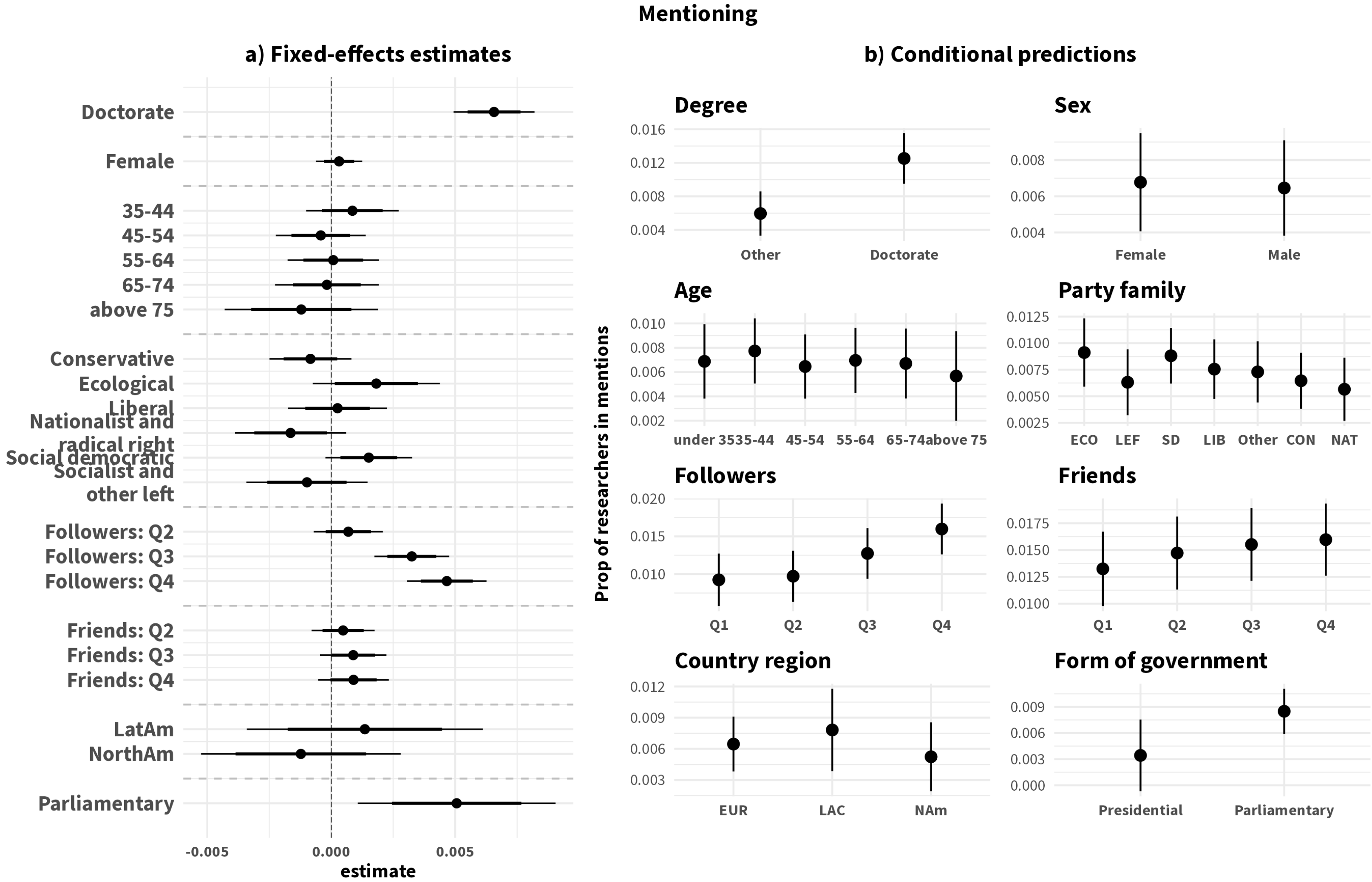



**Figure C1: Estimated effects of legislator and legislature characteristics on the proportion of mentions of researchers**

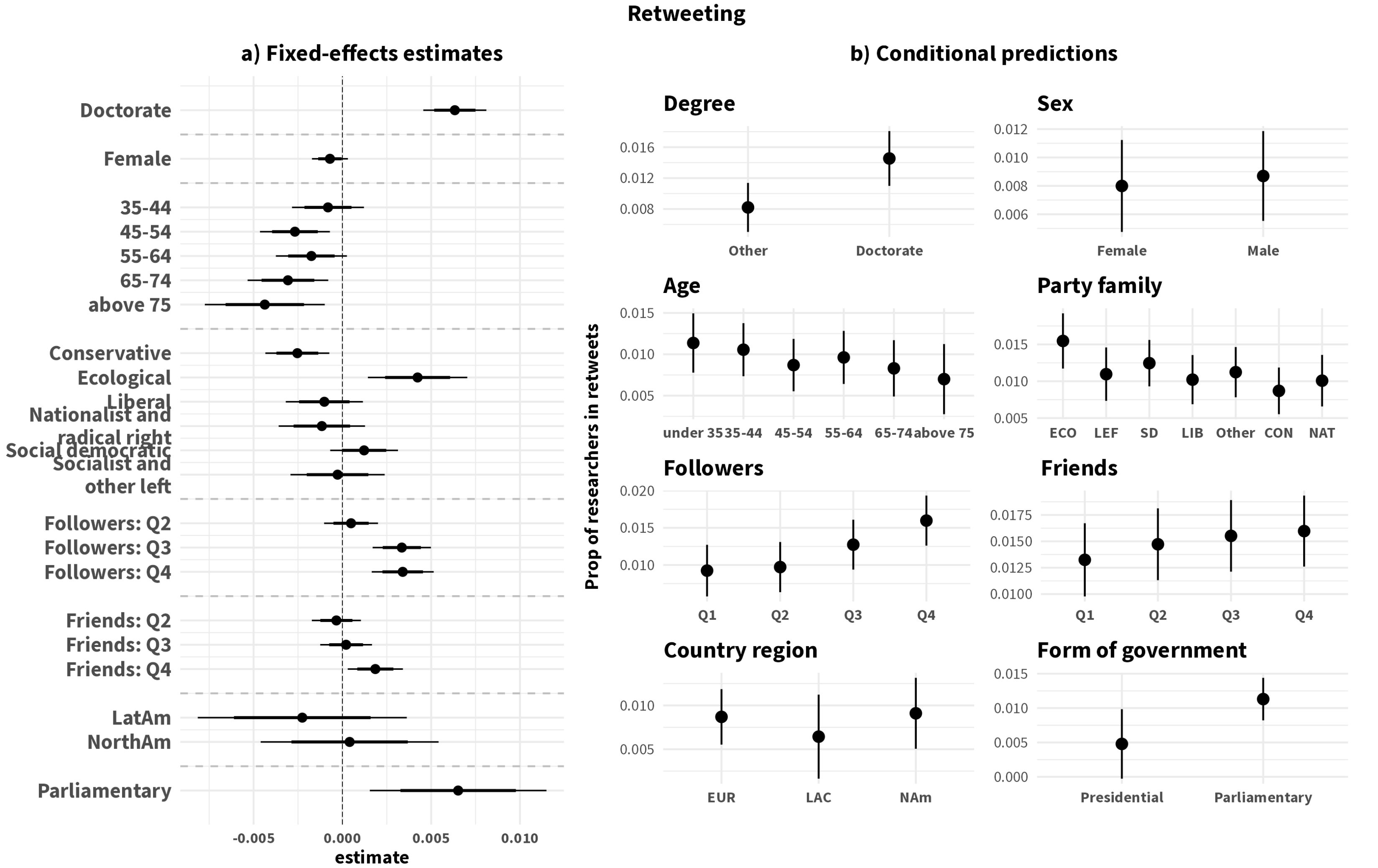



**Figure C2: Estimated effects of legislator and legislature characteristics on the proportion of retweeting of researchers**

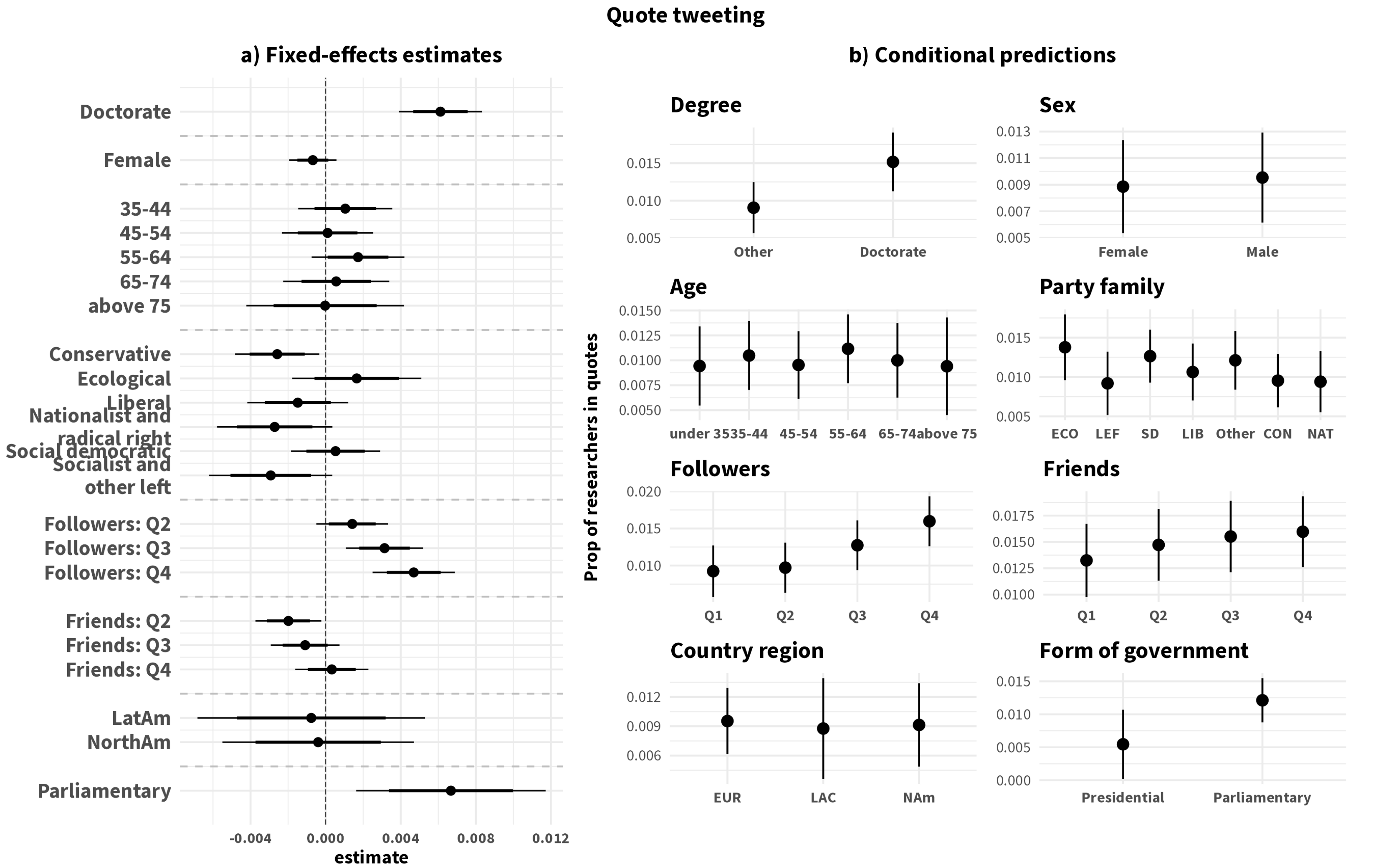



**Figure C3: Estimated effects of legislator and legislature characteristics on the proportion of quoting of researchers**

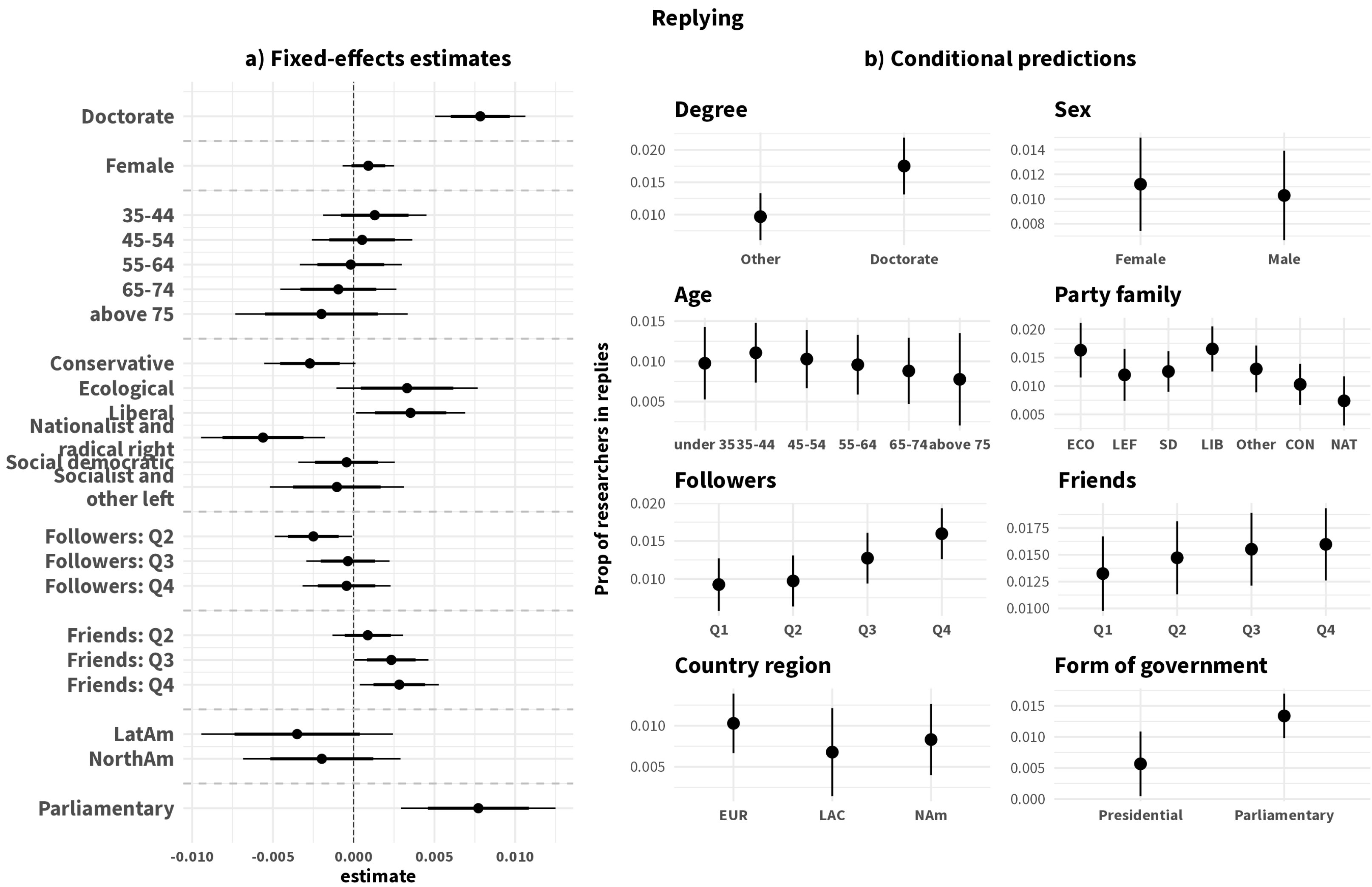



Figure C4: **Estimated effects of legislator and legislature characteristics on the proportion of replies to researchers**

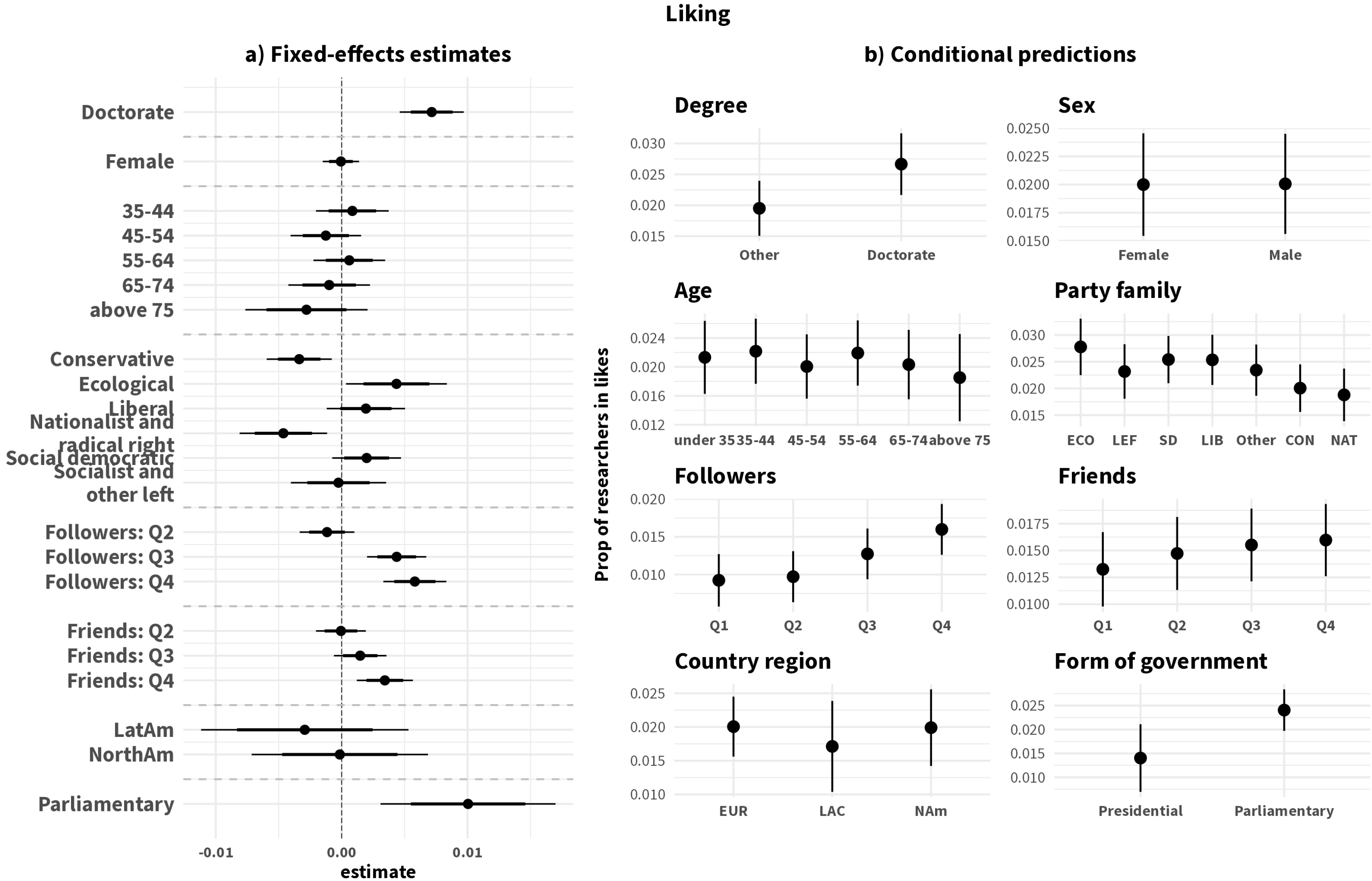



**Figure C5: Estimated effects of legislator and legislature characteristics on the proportion of likes of researcher Tweets**

## C.2 Estimated effects of legislator and legislature characteristics on the absolute number of researchers on behaviors

These are results from linear mixed-effects models with legislature random effects with age (under 35), party family (other), country region (Europe), system (presidential), and Q1 for followers and friends as references for categorical variables. Panel a presents the coefficients with 80% and 95% confidence intervals. The conditional predictions are computed with numeric covariates are held at their means and the other covariates at their modes: no research degree, presidential, European, male, 45-54, Conservative party, and the first quartile of followers and friends.

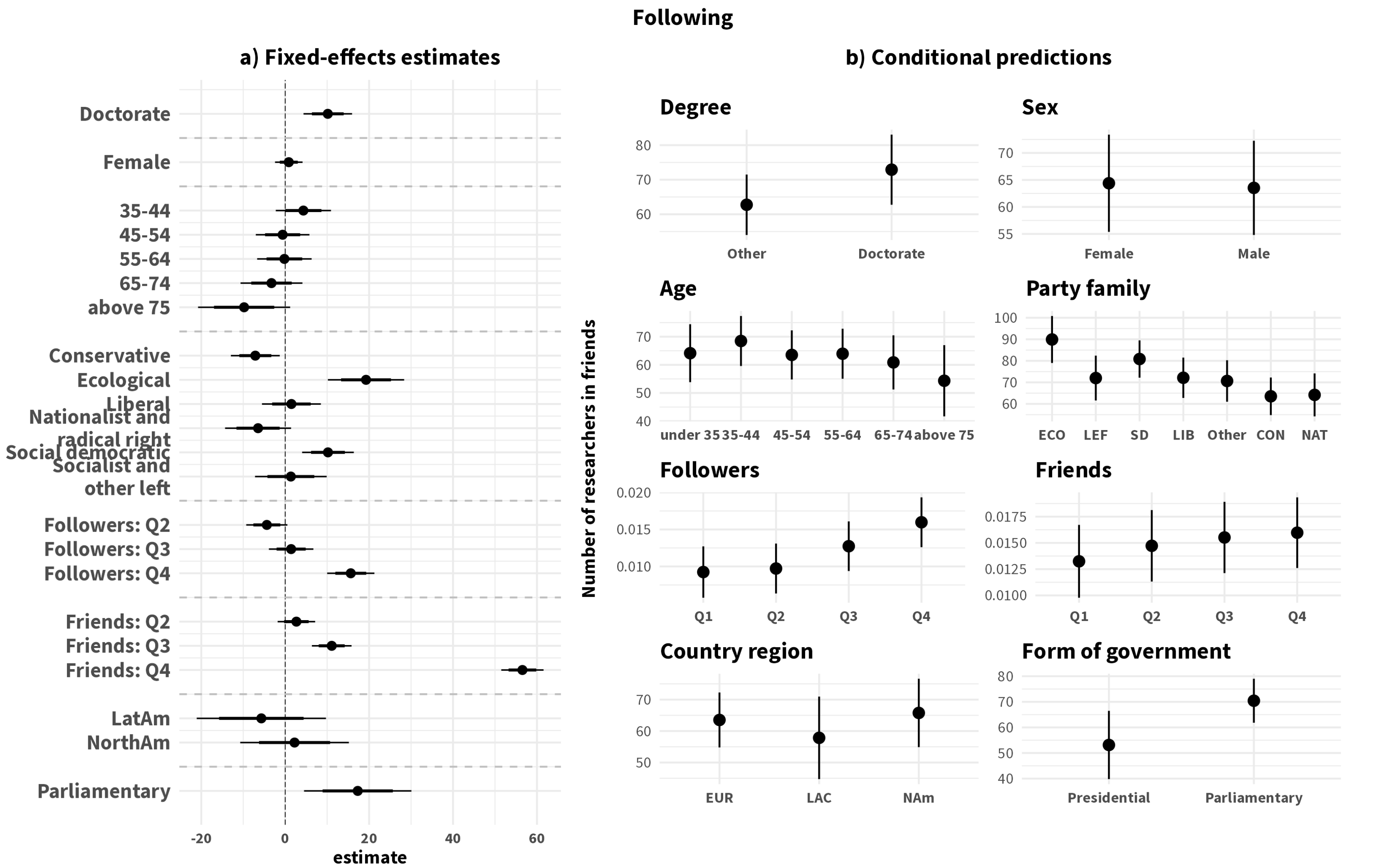



**Figure C6: Estimated effects of legislator and legislature characteristics on the absolute number of friends of researcher Tweets**

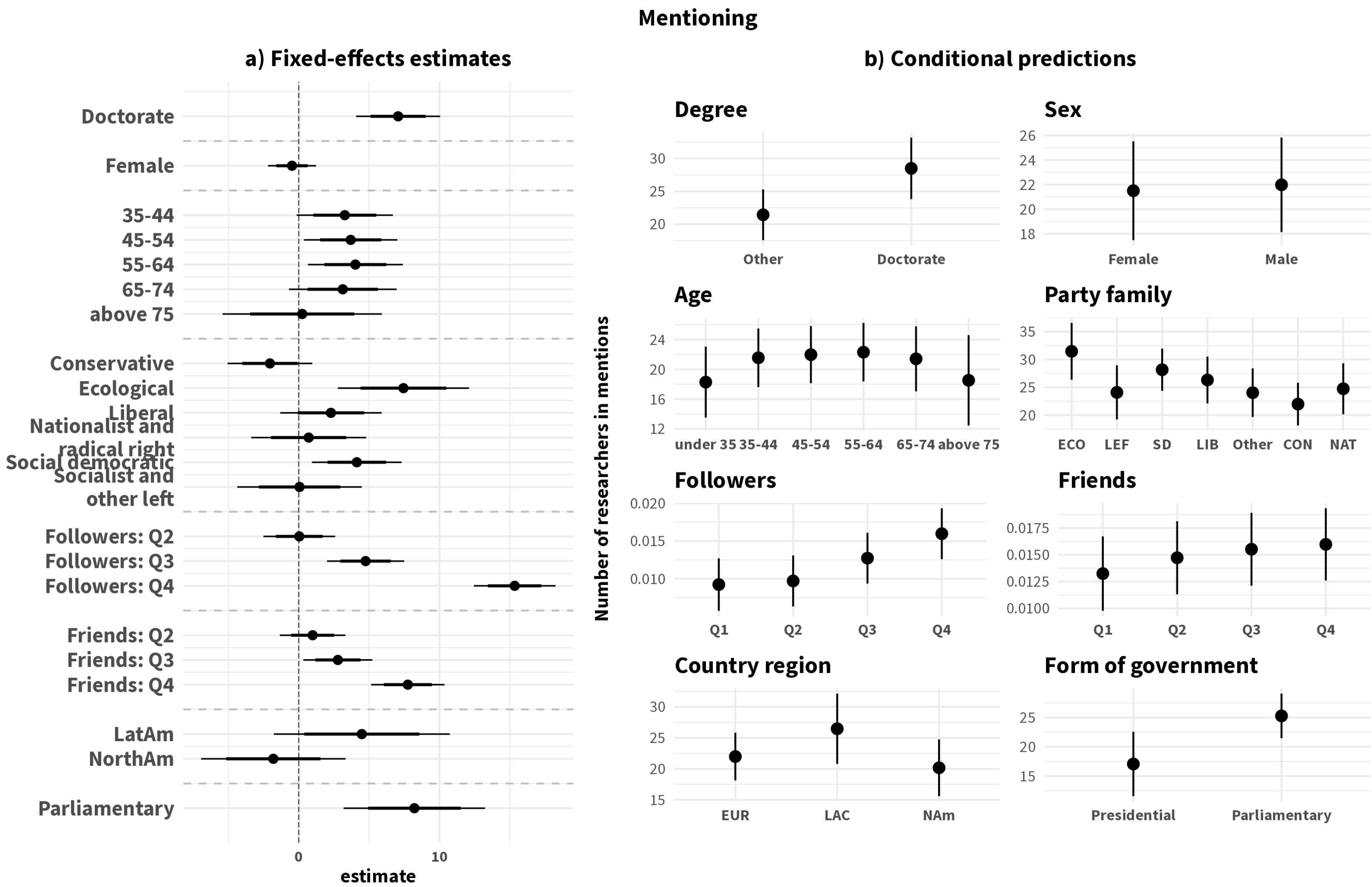



**Figure C7: Estimated effects of legislator and legislature characteristics on the absolute number of mentions of researchers**

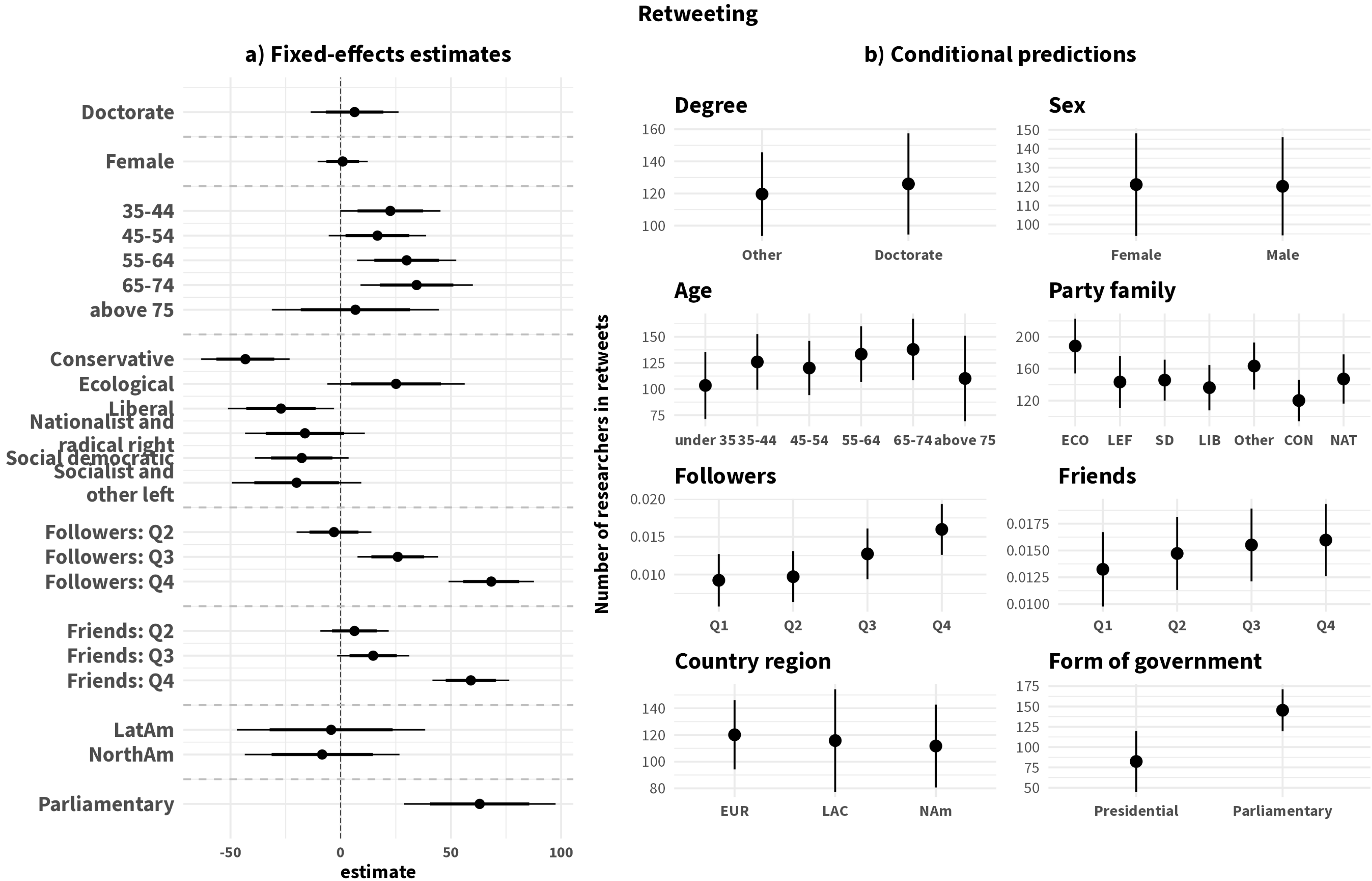



Figure C8: **Estimated effects of legislator and legislature characteristics on the absolute number of retweeting of researchers**

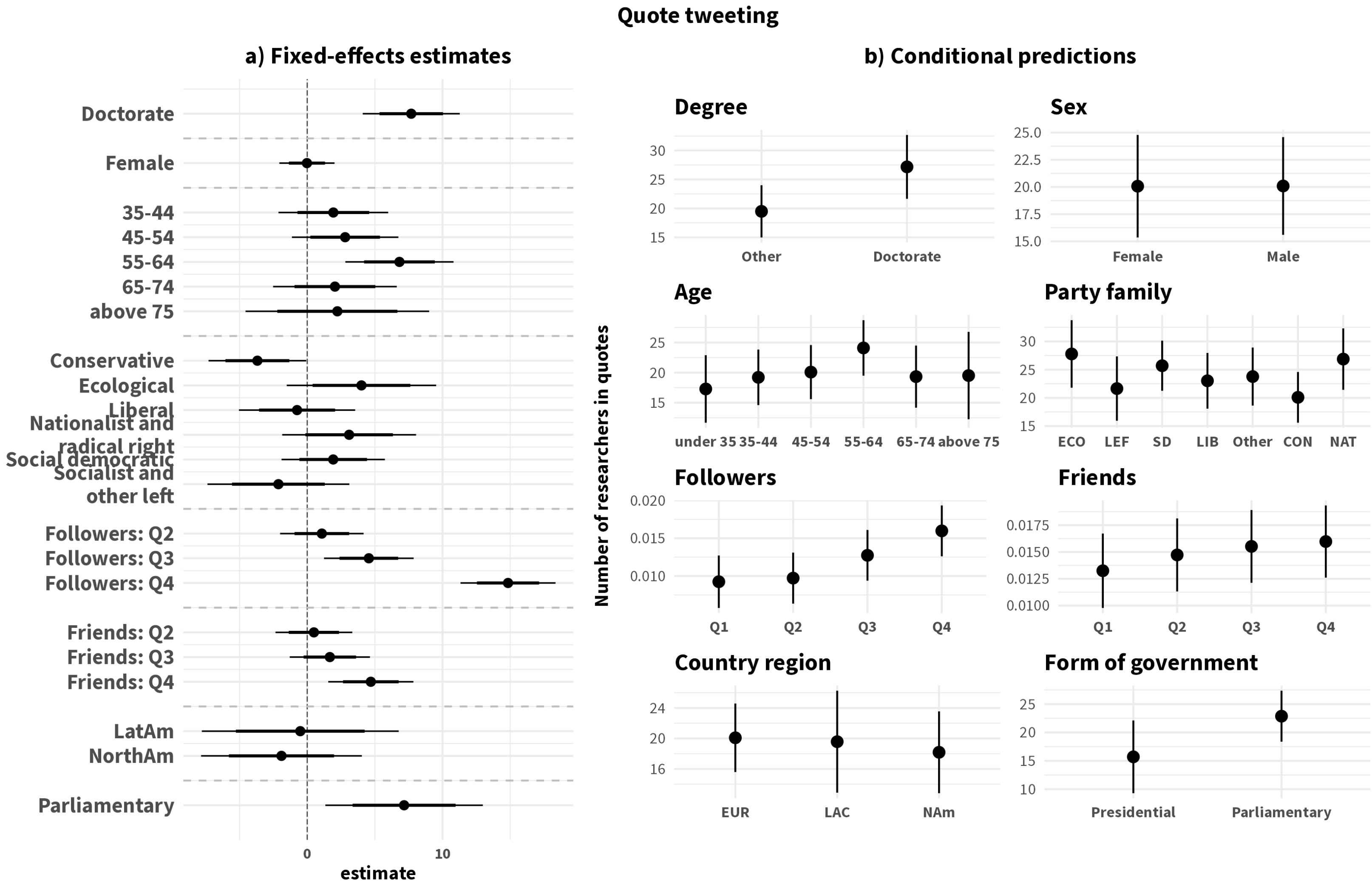



Figure C9: **Estimated effects of legislator and legislature characteristics on the absolute number of quoting of researchers**

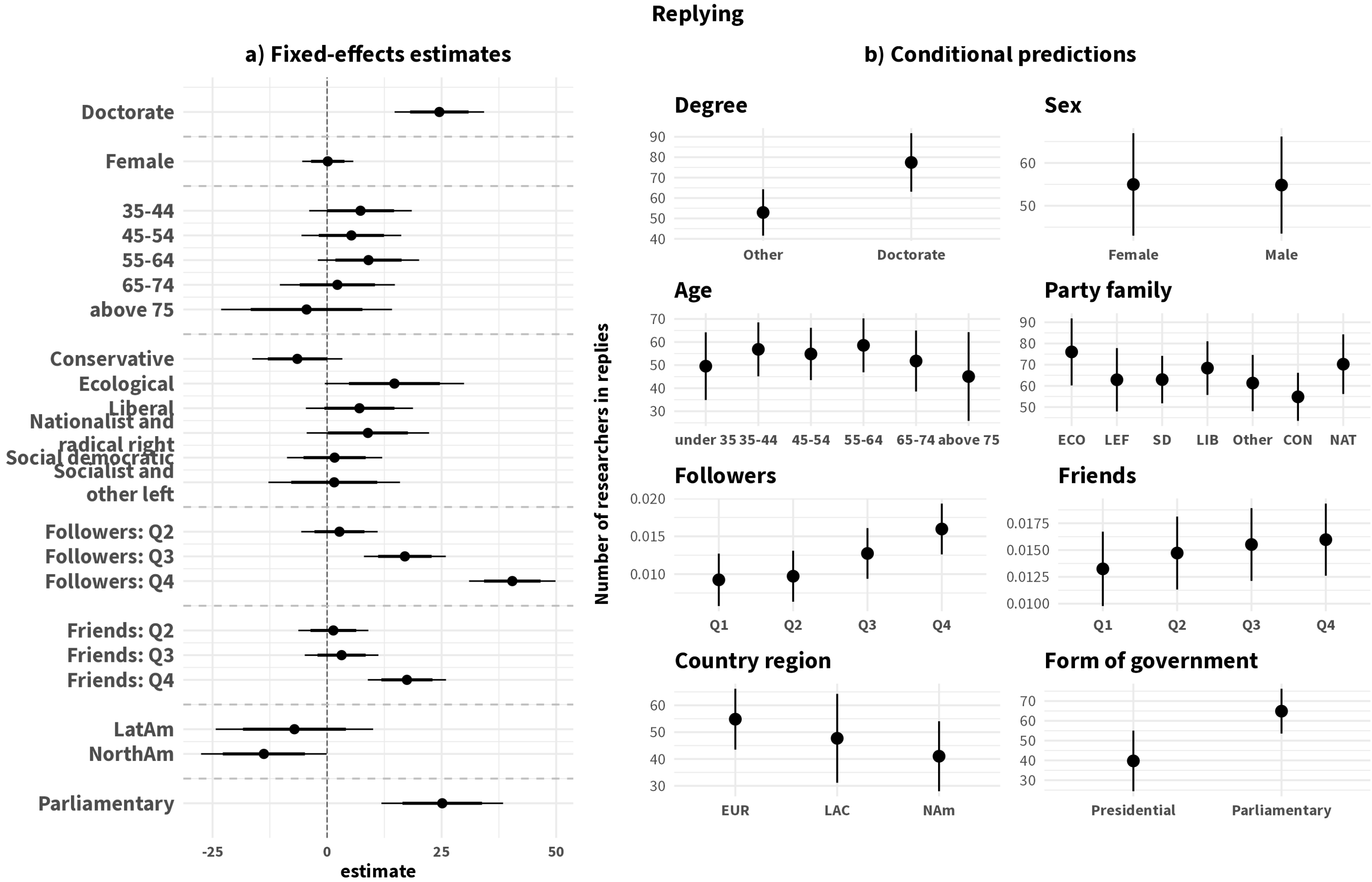



**Figure C10: Estimated effects of legislator and legislature characteristics on the absolute number of replies to researchers**

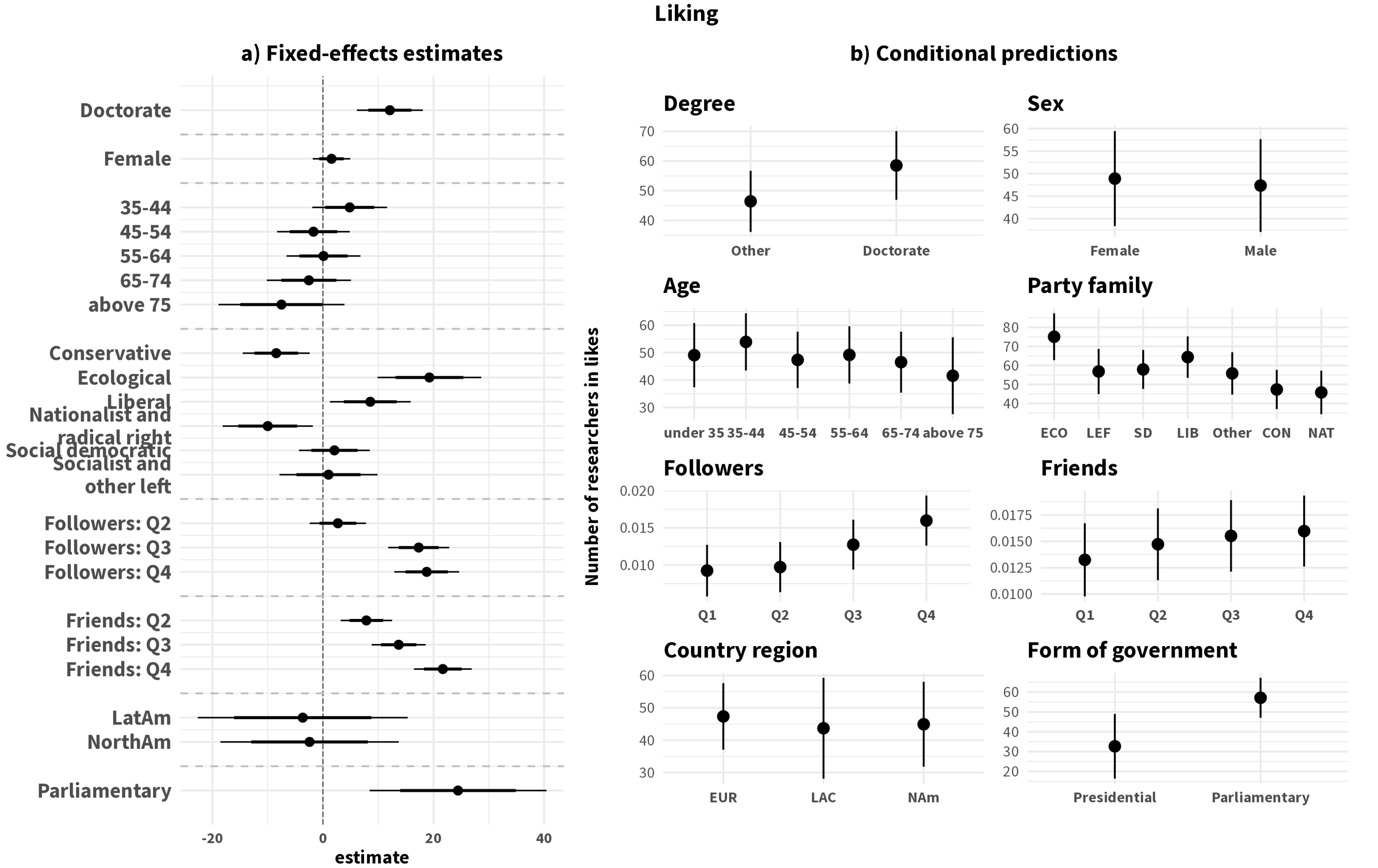



**Figure C11: Estimated effects of legislator and legislature characteristics on the absolute number of likes of researcher Tweets**

**Table C1:** Estimated effects of legislator and legislature characteristics on the absolute number of researchers across behaviors. (*Figs. C6-C11*)

| | Following | Mentioning | Retweeting | Quote tweeting | Replying | Liking |
|---|---|---|---|---|---|---|
| Doctoral studies | 10.073*** | 7.076*** | 6.422 | 7.708*** | 24.598*** | 12.123*** |
| | [4.286, 15.861] | [4.086, 10.065] | [−13.515, 26.359] | [4.121, 11.294] | [14.828, 34.368] | [6.156, 18.090] |
| Female | 0.825 | −0.489 | 0.902 | −0.023 | 0.164 | 1.561 |
| | [−2.457, 4.107] | [−2.196, 1.219] | [−10.454, 12.258] | [−2.062, 2.016] | [−5.415, 5.743] | [−1.828, 4.951] |
| Age: 35-44 (ref: Under 35) | 4.271 | 3.278+ | 22.643+ | 1.943 | 7.352 | 4.869 |
| | [−2.312, 10.853] | [−0.144, 6.700] | [−0.121, 45.406] | [−2.111, 5.997] | [−3.848, 18.552] | [−1.909, 11.647] |
| 45-54 | −0.592 | 3.705* | 16.737 | 2.817 | 5.316 | −1.708 |
| | [−6.988, 5.804] | [0.381, 7.029] | [−5.389, 38.862] | [−1.118, 6.752] | [−5.574, 16.205] | [−8.285, 4.868] |
| 55-64 | −0.183 | 4.044* | 30.025** | 6.837*** | 9.073 | 0.112 |
| | [−6.679, 6.313] | [0.668, 7.421] | [7.546, 52.504] | [2.837, 10.836] | [−1.987, 20.133] | [−6.574, 6.799] |
| 65-74 | −3.006 | 3.175 | 34.411** | 2.074 | 2.162 | −2.589 |
| | [−10.382, 4.371] | [−0.654, 7.004] | [8.929, 59.893] | [−2.494, 6.642] | [−10.397, 14.721] | [−10.195, 5.017] |
| above 75 | −9.185 | 0.334 | 6.518 | 2.275 | −4.675 | −7.619 |
| | [−20.149, 1.779] | [−5.321, 5.989] | [−31.346, 44.382] | [−4.508, 9.059] | [−23.316, 13.966] | [−18.987, 3.748] |
| Party family: Conservative (ref: Other) | −7.146* | −2.035 | −43.183*** | −3.705* | −6.479 | −8.457** |
| | [−12.976, −1.316] | [−5.051, 0.982] | [−63.299, −23.068] | [−7.319, −0.092] | [−16.326, 3.368] | [−14.523, −2.392] |
| Ecological | 19.246*** | 7.459** | 25.138 | 3.983 | 14.732+ | 19.198*** |
| | [10.148, 28.344] | [2.784, 12.133] | [−6.054, 56.331] | [−1.549, 9.515] | [−0.490, 29.954] | [9.793, 28.603] |
| Liberal | 1.507 | 2.209 | −27.616* | −0.845 | 6.923 | 8.524* |
| | [−5.527, 8.541] | [−1.411, 5.829] | [−51.719, −3.512] | [−5.156, 3.467] | [−4.817, 18.663] | [1.208, 15.839] |
| Nationalistic and radical right | −6.485 | 0.696 | −16.318 | 3.071 | 8.947 | −9.978* |
| | [−14.359, 1.389] | [−3.395, 4.788] | [−43.489, 10.852] | [−1.883, 8.024] | [−4.426, 22.321] | [−18.145, −1.812] |
| Social democratic | 10.046** | 4.158* | −17.395 | 1.921 | 1.781 | 2.107 |
| | [3.880, 16.212] | [0.969, 7.347] | [−38.687, 3.897] | [−1.901, 5.744] | [−8.619, 12.180] | [−4.305, 8.519] |
| Socialist and other left | 1.292 | 0.013 | −20.264 | −2.198 | 1.525 | 1.012 |
| | [−7.261, 9.846] | [−4.424, 4.450] | [−49.663, 9.134] | [−7.455, 3.058] | [−12.875, 15.924] | [−7.880, 9.903] |
| Followers qtl: Q2 (ref: Q1) | −4.372+ | 0.033 | −3.102 | 1.070 | 2.651 | 2.696 |
| | [−9.281, 0.536] | [−2.519, 2.585] | [−20.100, 13.896] | [−2.018, 4.158] | [−5.700, 11.002] | [−2.406, 7.798] |
| Q3 | 1.270 | 4.762*** | 26.101** | 4.551** | 17.028*** | 17.354*** |
| | [−4.036, 6.576] | [2.015, 7.510] | [7.795, 44.407] | [1.234, 7.868] | [8.075, 25.981] | [11.829, 22.880] |
| Q4 | 16.008*** | 15.416*** | 68.273*** | 14.861*** | 40.305*** | 18.705*** |
| | [10.383, 21.633] | [12.510, 18.323] | [48.904, 87.642] | [11.346, 18.377] | [30.849, 49.762] | [12.848, 24.563] |
| Friends qtl: Q2 (ref: Q1) | 2.734 | 0.990 | 6.189 | 0.495 | 1.335 | 7.836*** |
| | [−1.737, 7.206] | [−1.342, 3.321] | [−9.335, 21.713] | [−2.345, 3.334] | [−6.310, 8.980] | [3.187, 12.484] |
| Q3 | 11.063*** | 2.765* | 14.677+ | 1.664 | 3.157 | 13.704*** |
| | [6.334, 15.792] | [0.310, 5.220] | [−1.697, 31.051] | [−1.301, 4.630] | [−4.884, 11.198] | [8.808, 18.599] |
| Q4 | 56.514*** | 7.732*** | 58.949*** | 4.691** | 17.403*** | 21.699*** |
| | [51.470, 61.557] | [5.121, 10.343] | [41.525, 76.373] | [1.542, 7.839] | [8.862, 25.944] | [16.471, 26.926] |
| Region: LAC (ref: EUR) | −5.018 | 4.128 | −7.721 | −0.800 | −7.743 | −3.969 |
| | [−21.107, 11.070] | [−2.531, 10.788] | [−51.970, 36.527] | [−8.546, 6.946] | [−25.776, 10.289] | [−24.545, 16.607] |
| NAm | 1.209 | −1.326 | −3.874 | −1.493 | −12.821+ | −1.972 |
| | [−12.668, 15.086] | [−6.949, 4.297] | [−41.194, 33.447] | [−7.999, 5.013] | [−27.591, 1.948] | [−19.943, 16.000] |
| Parliamentary republic | 18.249** | 7.784** | 58.836** | 6.787* | 24.293*** | 23.998** |
| | [4.546, 31.952] | [2.273, 13.295] | [22.273, 95.399] | [0.410, 13.164] | [9.974, 38.612] | [6.166, 41.830] |
| SD (Observations) | 44.711 | 23.003 | 153.695 | 26.939 | 75.272 | 45.639 |
| Num.Obs. | 3381 | 3318 | 3335 | 3145 | 3311 | 3295 |
| R2 Marg. | 0.310 | 0.139 | 0.107 | 0.086 | 0.095 | 0.163 |
| R2 Cond. | 0.330 | 0.154 | 0.122 | 0.101 | 0.103 | 0.203 |
| ICC | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| RMSE | 44.52 | 22.90 | 153.04 | 26.82 | 74.96 | 45.43 |

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

## C.3 Auxiliary analyses for pre- and during COVID legislator-researcher engagement comparison

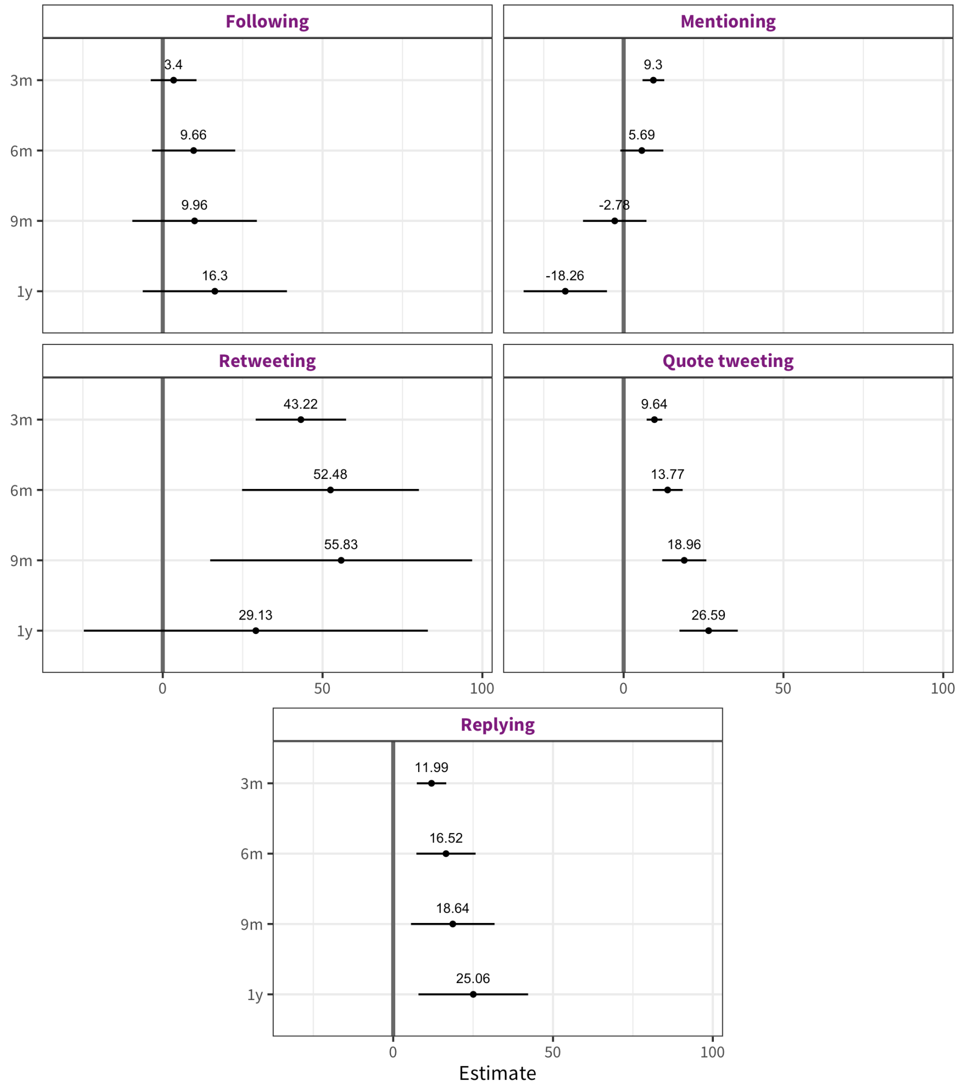



**Figure C12: Differences-in-means between pre- and during-COVID number of behaviors.** Results from OLS estimates

**Table C2: Differences-in-means between pre- and during-COVID number of behaviors across bandwidths**. Results from OLS estimates (*Fig. C12*)

| Behavior | Estimate | Std. error | P. value | Conf. int |
|---|---|---|---|---|
| Following (3m) | 3.48 | 3.66 | 0.34 | [-3.7,10.7] |
| Following (6m) | 9.79 | 6.65 | 0.14 | [-3.3,22.8] |
| Following (9m) | 10.08 | 9.96 | 0.31 | [-9.5,29.6] |
| Following (1y) | 16.44 | 11.53 | 0.15 | [-6.2,39] |
| Mentioning (3m) | 10.23 | 1.75 | 0.00 | [6.8,13.7] |
| Mentioning (6m) | 7.37 | 3.47 | 0.03 | [0.6,14.2] |
| Mentioning (9m) | -0.39 | 5.12 | 0.94 | [-10.4,9.6] |
| Mentioning (1y) | -15.25 | 6.73 | 0.02 | [-28.4,-2.1] |
| Retweeting (3m) | 46.42 | 7.29 | 0.00 | [32.1,60.7] |
| Retweeting (6m) | 58.35 | 14.28 | 0.00 | [30.4,86.3] |
| Retweeting (9m) | 64.08 | 21.15 | 0.00 | [22.6,105.5] |
| Retweeting (1y) | 39.47 | 27.78 | 0.16 | [-15,93.9] |
| Quote tweeting (3m) | 10.12 | 1.25 | 0.00 | [7.7,12.6] |
| Quote tweeting (6m) | 14.67 | 2.42 | 0.00 | [9.9,19.4] |
| Quote tweeting (9m) | 20.28 | 3.56 | 0.00 | [13.3,27.3] |
| Quote tweeting (1y) | 28.39 | 4.69 | 0.00 | [19.2,37.6] |
| Replying (3m) | 12.74 | 2.38 | 0.00 | [8.1,17.4] |
| Replying (6m) | 17.90 | 4.77 | 0.00 | [8.5,27.3] |
| Replying (9m) | 20.59 | 6.75 | 0.00 | [7.4,33.8] |
| Replying (1y) | 27.66 | 8.85 | 0.00 | [10.3,45] |

**Table C3:** Results from logistic regression on following and engagement with academic researchers during the COVID versus pre-COVID periods with a 12 week bandwidth with party family interaction (*Fig. C13*)

| | Following | Mentioning | Retweeting | Quote tweeting | Replying |
|---|---|---|---|---|---|
| Post-COVID | $0.391^{***}$ | $0.362^{**}$ | $0.399^{***}$ | $0.185^{+}$ | $-0.079$ |
| | [0.287, 0.495] | [0.113, 0.615] | [0.308, 0.491] | [−0.019, 0.393] | [−0.225, 0.067] |
| Ecological | $1.026^{***}$ | $1.420^{***}$ | $1.478^{***}$ | $0.960^{***}$ | $1.037^{***}$ |
| | [0.892, 1.159] | [1.077, 1.750] | [1.359, 1.597] | [0.656, 1.252] | [0.871, 1.201] |
| Liberal | $0.194^{**}$ | $0.663^{***}$ | $0.556^{***}$ | $0.017$ | $0.305^{***}$ |
| | [0.066, 0.320] | [0.369, 0.954] | [0.429, 0.681] | [−0.300, 0.319] | [0.131, 0.477] |
| Nationalist and radical right | $-0.973^{***}$ | $-1.017^{*}$ | $-0.391^{***}$ | $0.260$ | $-0.932^{***}$ |
| | [−1.300, −0.674] | [−2.201, −0.148] | [−0.587, −0.205] | [−0.077, 0.579] | [−1.320, −0.581] |
| Other | $0.274^{***}$ | $0.749^{***}$ | $0.975^{***}$ | $0.832^{***}$ | $0.146$ |
| | [0.132, 0.412] | [0.437, 1.055] | [0.877, 1.073] | [0.589, 1.073] | [−0.040, 0.328] |
| Social democratic | $0.618^{***}$ | $1.042^{***}$ | $1.101^{***}$ | $0.577^{***}$ | $0.419^{***}$ |
| | [0.525, 0.712] | [0.814, 1.277] | [1.014, 1.188] | [0.385, 0.774] | [0.291, 0.549] |
| Socialist and other left | $-0.034$ | $0.087$ | $0.049$ | $-0.255$ | $0.004$ |
| | [−0.207, 0.132] | [−0.364, 0.502] | [−0.139, 0.230] | [−0.742, 0.180] | [−0.254, 0.248] |
| Post-COVID*Ecological | $-0.108$ | $0.064$ | $-0.093$ | $0.066$ | $-0.043$ |
| | [−0.289, 0.073] | [−0.365, 0.497] | [−0.241, 0.055] | [−0.306, 0.443] | [−0.269, 0.183] |
| Post-COVID*Liberal | $0.141^{+}$ | $-0.078$ | $-0.307^{***}$ | $0.219$ | $0.232^{*}$ |
| | [−0.026, 0.308] | [−0.451, 0.296] | [−0.462, −0.150] | [−0.149, 0.596] | [0.009, 0.456] |
| Post-COVID*Nationalist and radical right | $-0.060$ | $-0.415$ | $-0.428^{***}$ | $-0.522^{*}$ | $-0.054$ |
| | [−0.460, 0.350] | [−1.877, 1.048] | [−0.676, −0.177] | [−0.973, −0.072] | [−0.560, 0.456] |
| Post-COVID*Other | $0.121$ | $0.070$ | $-0.130^{*}$ | $-0.227$ | $0.130$ |
| | [−0.060, 0.302] | [−0.329, 0.471] | [−0.253, −0.006] | [−0.551, 0.096] | [−0.119, 0.380] |
| Post-COVID*Social democratic | $-0.219^{***}$ | $-0.164$ | $-0.383^{***}$ | $0.070$ | $-0.100$ |
| | [−0.347, −0.092] | [−0.463, 0.133] | [−0.492, −0.274] | [−0.178, 0.316] | [−0.280, 0.079] |
| Post-COVID*Socialist and other left | $-0.212^{+}$ | $-0.199$ | $-0.092$ | $0.069$ | $0.004$ |
| | [−0.455, 0.031] | [−0.770, 0.380] | [−0.322, 0.141] | [−0.497, 0.659] | [−0.326, 0.338] |
| Num.Obs. | 334 670 | 315 542 | 925 499 | 166 371 | 266 846 |
| RMSE | 0.15 | 0.07 | 0.11 | 0.11 | 0.12 |

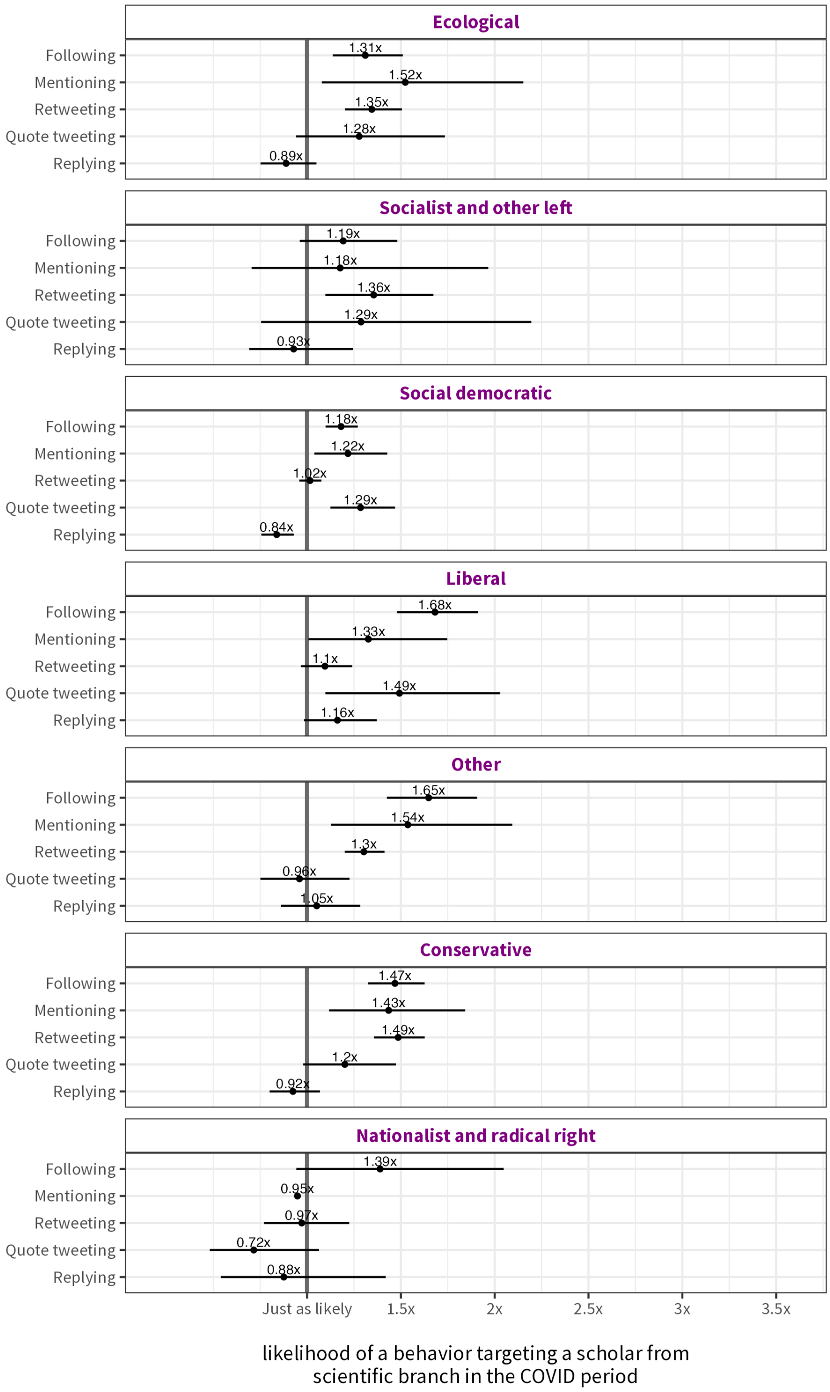
+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

**Table C4:** Results from a logistic mixed effects model with country of legislature random effects on following and engagement with academic researchers during the COVID versus pre-COVID periods with a 12 week bandwidth with Doctorate interaction (*Fig. C14*)

| | Following | Mentioning | Retweeting | Quote tweeting | Replying |
|---|---|---|---|---|---|
| Post-COVID | $0.313^{***}$ | $0.351^{***}$ | $0.177^{***}$ | $0.190^{***}$ | $-0.032$ |
| | [0.262, 0.363] | [0.242, 0.461] | [0.141, 0.214] | [0.094, 0.285] | [−0.101, 0.036] |
| Doctorate | $0.412^{***}$ | $0.667^{***}$ | $0.344^{***}$ | $0.534^{***}$ | $0.710^{***}$ |
| | [0.296, 0.529] | [0.456, 0.878] | [0.276, 0.412] | [0.324, 0.745] | [0.584, 0.836] |
| Post-COVID*Doctorate | $0.201^{**}$ | $-0.335^{*}$ | $-0.175^{***}$ | $-0.050$ | $-0.109$ |
| | [0.053, 0.348] | [−0.606, −0.063] | [−0.259, −0.090] | [−0.308, 0.208] | [−0.272, 0.055] |
| Num.Obs. | 313 252 | 315 542 | 925 499 | 166 371 | 266 846 |
| R2 Marg. | 0.012 | 0.013 | 0.003 | 0.008 | 0.008 |
| R2 Cond. | 0.097 | 0.062 | 0.146 | 0.076 | 0.264 |
| ICC | 0.1 | 0.0 | 0.1 | 0.1 | 0.3 |
| RMSE | 0.15 | 0.07 | 0.11 | 0.11 | 0.12 |

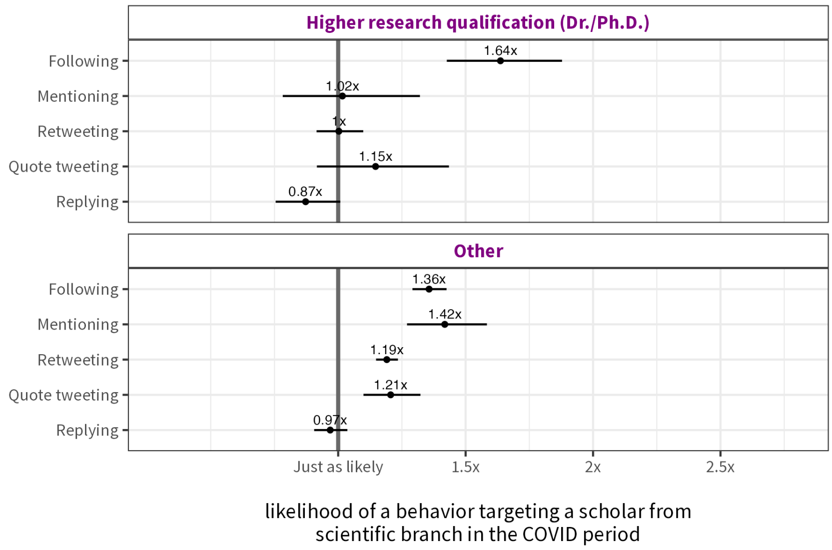
+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

**Figure C13: Subgroup marginal effects on following and engagement with academic researchers during the COVID versus pre-COVID periods with a ±12 week bandwidth (Party family).**




**Note.** Results from logistic regression model. The estimates in the figure are relative risks representing the ratio of the probability of an event in the COVID period to the probability of an outcome in a pre-COVID period.

**Figure C14: Subgroup marginal effects on following and engagement with academic researchers during the COVID versus pre-COVID periods with a ±12 week bandwidth (Research qualifications)**.




**Note.** Results from a logistic mixed-effects models with country of legislature random effects. The estimates in the figure are relative risks representing the ratio of the probability of an event in the COVID period to the probability of an outcome in a pre-COVID period.

## Appendix D Software statement

The computer code and processed data for this study is available at XXX. I used R version 4.2.1 (R Core Team, 2022) and the following R packages:

`academictwitteR` v. 0.3.1 (Barrie and Ho, 2021)
`data.table` v. 1.14.2 (Dowle and Srinivasan, 2021)
`estimatr` v. 1.0.0 (Blair et al., 2022)
`ggh4x` v. 0.2.6 (van den Brand, 2023)
`gt` v. 0.9.0 (Iannone et al., 2023)
`gtExtras` v. 0.5.0.9000 (Mock, 2023)
`janitor` v. 2.1.0 (Firke, 2021)
`kableExtra` v. 1.3.4 (Zhu, 2021)
`lme4` v. 1.1.31 (Bates, Mächler, et al., 2015)
`marginaleffects` v. 0.11.1 (Arel-Bundock, 2023)
`Matrix` v. 1.5.3 (Bates, Maechler, and Jagan, 2022)
`modelsummary` v. 1.4.3 (Arel-Bundock, 2022)
`pacman` v. 0.5.1 (Rinker and Kurkiewicz, 2018)
`patchwork` v. 1.1.2 (Pedersen, 2022)
`psych` v. 2.2.5 (Revelle, 2022)
`reactable` v. 0.4.4 (Lin, 2023)
`reactablefmtr` v. 2.0.0 (Cuilla, 2022)
`rtweet` v. 1.0.2 (Kearney, 2019b)
`scales` v. 1.3.0 (Wickham, Pedersen, and Seidel, 2023)
`tidyverse` v. 2.0.0 (Wickham, Averick, et al., 2019)
`urltools` v. 1.7.3 (Keyes et al., 2019)
`webshot2` v. 0.1.0 (Chang, 2022)
`WikidataR` v. 2.3.3 (Shafee, Keyes, and Signorelli, 2021)
`xtable` v. 1.8.4 (Dahl et al., 2019)

## References


Arel-Bundock, Vincent (2022). “modelsummary: Data and Model Summaries in R”. *Journal of Statistical Software* 103.1, pp. 1–23.

— (2023). *marginaleffects: Predictions, Comparisons, Slopes, Marginal Means, and Hypothesis Tests*.

Barrie, Christopher and Justin Chun-ting Ho (2021). “academictwitteR: an R package to access the Twitter Academic Research Product Track v2 API endpoint”. *Journal of Open Source Software* 6.62, p. 3272.

Bates, Douglas, Martin Mächler, Ben Bolker, and Steve Walker (2015). “Fitting Linear Mixed-Effects Models Using lme4”. *Journal of Statistical Software* 67.1, pp. 1–48.

Bates, Douglas, Martin Maechler, and Mikael Jagan (2022). *Matrix: Sparse and Dense Matrix Classes and Methods*.

Blair, Graeme, Jasper Cooper, Alexander Coppock, Macartan Humphreys, and Luke Sonnet (2022). *estimatr: Fast Estimators for Design-Based Inference*.

Chang, Winston (2022). *webshot2: Take Screenshots of Web Pages*.

Cuilla, Kyle (2022). *reactablefmtr: Streamlined Table Styling and Formatting for Reactable*.

Dahl, David B., David Scott, Charles Roosen, Arni Magnusson, and Jonathan Swinton (2019). *xtable: Export Tables to LaTeX or HTML*.

Dowle, Matt and Arun Srinivasan (2021). *data.table: Extension of ‘data.frame’*.

Firke, Sam (2021). *janitor: Simple Tools for Examining and Cleaning Dirty Data*.

Göbel, Sascha and Simon Munzert (2022). “The Comparative Legislators Database”. *British Journal of Political Science* 52.3, pp. 1398–1408.

“International Standard Classification of Education, ISCED 1997” (2003). In: *Advances in Cross-National Comparison: A European Working Book for Demographic and Socio-Economic Variables*. Ed. by Jürgen H. P. Hoffmeyer-Zlotnik and Christof Wolf. Boston, MA: Springer US, pp. 195–220.

Iannone, Richard, Joe Cheng, Barret Schloerke, Ellis Hughes, Alexandra Lauer, and JooYoung Seo (2023). *gt: Easily Create Presentation-Ready Display Tables*.

Kearney, Michael W. (2019a). “rtweet: Collecting and analyzing Twitter data”. *Journal of Open Source Software* 4.42, p. 1829.

— (2019b). “rtweet: Collecting and analyzing Twitter data”. *Journal of Open Source Software* 4.42, p. 1829.

Keyes, Os, Jay Jacobs, Drew Schmidt, Mark Greenaway, Bob Rudis, Alex Pinto, Maryam Khezrzadeh, Peter Meilstrup, Adam M. Costello, Jeff Bezanson, Peter Meilstrup, and Xueyuan Jiang (2019). *urltools: Vectorised Tools for URL Handling and Parsing*.

Lehmann, Pola, Simon Franzmann, Tobias Burst, Sven Regel, Felicia Riethmüller, Andrea Volkens, Bernhard Weßels, and Lisa Zehnter (2023). *The Manifesto Data Collection. Manifesto Project (MRG/CMP/MARPOR). Version 2023a*.

Lin, Greg (2023). *reactable: Interactive Data Tables for R*.

Meeder, Brendan, Brian Karrer, Amin Sayedi, R. Ravi, Christian Borgs, and Jennifer Chayes (2011). “We know who you followed last summer: inferring social link creation times in twitter”. In: *Proceedings of the 20th international conference on World wide web*. WWW ’11. New York, NY, USA: Association for Computing Machinery, pp. 517–526.

Mock, Thomas (2023). *gtExtras: Extending ‘gt’ for Beautiful HTML Tables*.

Mongeon, Philippe, Timothy D. Bowman, and Rodrigo Costas (2023). “An open data set of scholars on Twitter”. *Quantitative Science Studies*, pp. 1–11.

Pedersen, Thomas Lin (2022). *patchwork: The Composer of Plots*.

Priem, Jason, Heather Piwowar, and Richard Orr (2022). *OpenAlex: A fully-open index of scholarly works, authors, venues, institutions, and concepts*.

R Core Team (2022). *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. Vienna, Austria.

Revelle, William (2022). *psych: Procedures for Psychological, Psychometric, and Personality Research*. Northwestern University. Evanston, Illinois.

Rinker, Tyler W. and Dason Kurkiewicz (2018). *pacman: Package Management for R*. Buffalo, New York.

Shafee, Thomas, Os Keyes, and Serena Signorelli (2021). *WikidataR: Read-Write API Client Library for Wikidata*.

Shen, Zhihong, Hao Ma, and Kuansan Wang (2018). *A Web-scale system for scientific knowledge exploration*.

van den Brand, Teun (2023). *ggh4x: Hacks for 'ggplot2'*.
Wickham, Hadley, Mara Averick, Jennifer Bryan, Winston Chang, Lucy D'Agostino McGowan, Romain François, Garrett Grolemund, Alex Hayes, Lionel Henry, Jim Hester, Max Kuhn, Thomas Lin Pedersen, Evan Miller, Stephan Milton Bache, Kirill Müller, Jeroen Ooms, David Robinson, Dana Paige Seidel, Vitalie Spinu, Kohske Takahashi, Davis Vaughan, Claus Wilke, Kara Woo, and Hiroaki Yutani (2019). "Welcome to the tidyverse". *Journal of Open Source Software* 4.43, p. 1686.
Wickham, Hadley, Thomas Lin Pedersen, and Dana Seidel (2023). *scales: Scale Functions for Visualization*.
Wojcieszak, Magdalena, Andreu Casas, Xudong Yu, Jonathan Nagler, and Joshua A. Tucker (2022). "Most users do not follow political elites on Twitter; those who do show overwhelming preferences for ideological congruity". *Science Advances* 8.39, eabn9418.
Zhu, Hao (2021). *kableExtra: Construct Complex Table with 'kable' and Pipe Syntax*.